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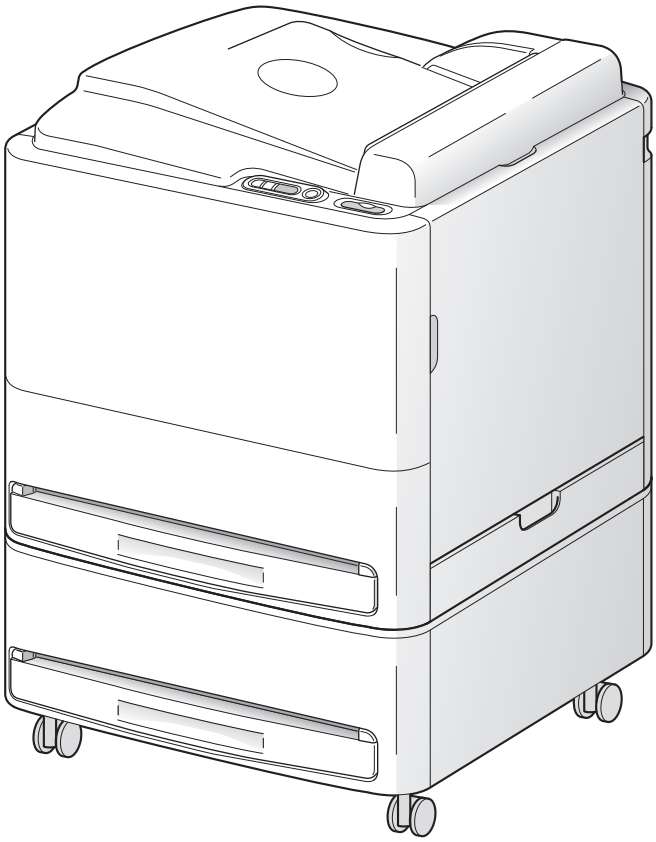
Fuji Medical Dry Laser Imager

DRYPIX Smart

(DRYPIX 6000)

Reference Guide

8th Edition : February 2017



For Safe Operation

Product Overview

Basic Operation

Utility

Troubleshooting

Care and Maintenance

Appendix

Maintenance and Inspection

This Manual describes details on how to operate the DRYPIX Smart and cautions to be observed when operating it. Please read this manual thoroughly before actually operating the DRYPIX Smart After reading this manual, store it nearby the DRYPIX Smart so that you can see it whenever necessary.

FUJIFILM Corporation

897N102437G

Introduction

This manual provides detailed explanations of operation methods and cautions to promote proper understanding of functions and more effective usage.

Accompanying documents were originally drafted in the English language.

Installation may only be conducted by authorized service personal.

This manual is applicable to the following software.

- DRYPIX Smart Main Software V. 1.0 or later

Indications for use (for U.S.)

Fuji Medical Dry Laser Imager, model DRYPIX Smart (DRYPIX 6000) is indicated for use in providing diagnostic quality medical images on film for aid in physician diagnosis, including the printing of images and associated identification information from various digital imaging source modalities, including but not limited to,Computed Tomography (CT), Magnetic Resonance Imaging (MRI), Ultrasound,Computed Radiography, Digital Radiography, Digital Mammography and Nuclear Medicine.

Intended use (for European Union and other countries.)

Fuji Medical Dry Laser Imager DRYPIX Smart (DRYPIX 6000) is a device that prints digital image data transferred via the DICOM network from the FCR/FDR Image Reader or DR, CT, MRI and other imaging modalities onto the Fuji Medical Dry Laser Image Film.

Note : The above statements were determined by applicable medical device regulations which vary throughout the world. These statements are subject to revision when additional clearance or approval is obtained.

⚠ CAUTIONS

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2. **The information contained in this manual may be subject to change without prior notice.**
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7. **FUJIFILM Corporation shall not be liable for malfunctions and damages resulting from use under environment conditions outside the range of using conditions for this product such as power supply, installation environment, etc. contained in this manual.**
8. **FUJIFILM Corporation shall not be liable for malfunctions and damages resulting from natural disasters such as fires, earthquakes, floods, lightning, etc.**

Caution : Rx Only in the United States (Federal law restricts this device to sale by or on the order of a physician.)

Open-Source Software Used in This Product

This product uses third party's software that is made available as open source software or free software.
SUCH SOFTWARE IS PROVIDED "AS IS" WITH NO WARRANTY OF ANY KIND AS TO ITS MERCHANTABILITY OR FITNESS FOR ANY PARTICULAR PURPOSE.
For information on open source software used in this product, please see the attached CD.
Source codes for certain type of open source software used in this product are available at delivery cost. If you would like to receive such source codes, please contact FUJIFILM dealer or the service representatives at the agency from which you purchased this product.
(Please be noted that any inquiries concerning the contents of source codes should be directed to original licensers of open source software.)

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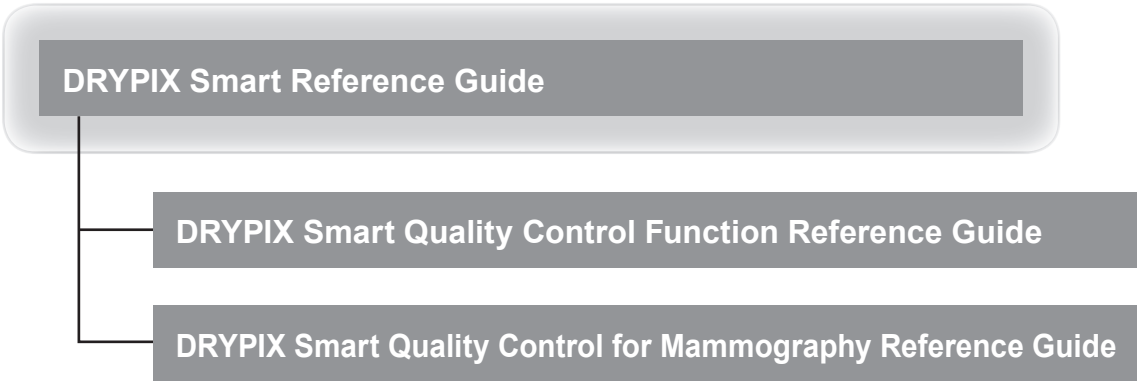
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DRYPIX Smart Manuals



Contents at a Glance

Chapter 1 **For Safe Operation**
This chapter presents Warnings and Cautions we wish you to observe for the safe operation of this equipment.

Chapter 2 **Product Overview**
This chapter presents overview and major features of this equipment.

Chapter 3 **Basic Operation**
This chapter describes routine operation procedures, including how to start up/ shut down this equipment and also how to replace a film pack.

Chapter 4 **Utility**
This chapter describes the film density calibration, date and time setting as well as other functions.

Chapter 5 **Troubleshooting**
This chapter explains how to troubleshoot an error.

Chapter 6 **Care and Maintenance**
This chapter describes daily care and maintenance we wish you to perform so that you can use the equipment optimally.

Appendix **Appendix A Specifications**
This chapter presents major specifications of this equipment.

Maintenance and Inspection

How to Read This Manual

Basic page configuration

Please have a good grasp of the basic page configuration of this Manual, as illustrated below, for you to use it more efficiently.

Section title
Shows the title of an operation procedure described in the section.

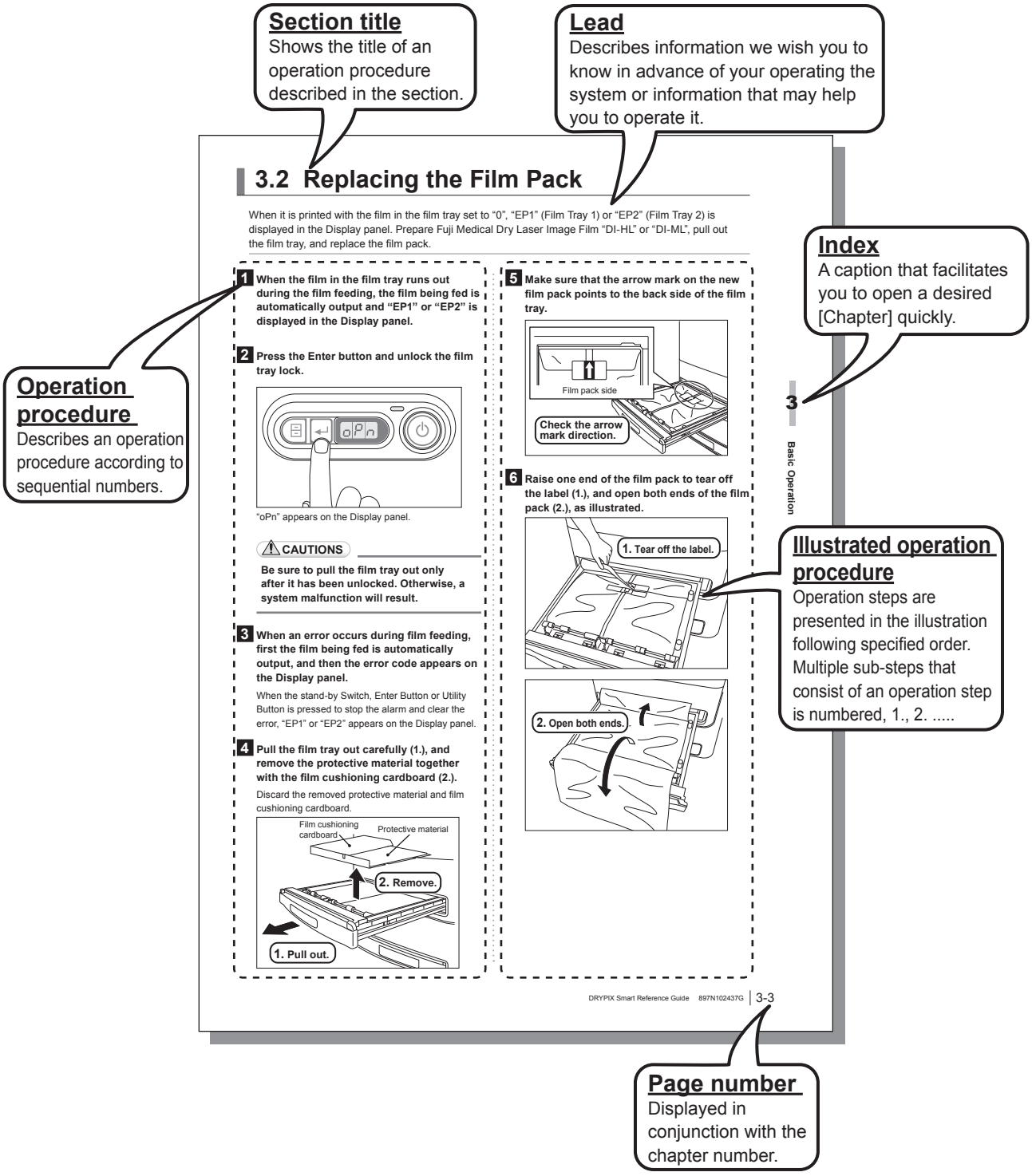
Lead
Describes information we wish you to know in advance of your operating the system or information that may help you to operate it.

Index
A caption that facilitates you to open a desired [Chapter] quickly.

Operation procedure
Describes an operation procedure according to sequential numbers.

Illustrated operation procedure
Operation steps are presented in the illustration following specified order. Multiple sub-steps that consist of an operation step is numbered, 1., 2. .....

Page number
Displayed in conjunction with the chapter number.



Marks

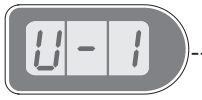
Information items to be observed when you are operating this system and the supplementary remarks are described in this manual with the respective marks.
For the safe system operation, be sure to observe **Warning/Caution**.

WARNING	**Indicates hazardous situations that may lead to serious injury or even death if the precaution is not or could not be followed.**
CAUTIONS	**Indicates hazardous situations that may lead to mild or moderate injury or physical damages if the caution is not or cannot be followed.**
	Indicates information that should be noted.
HINT	Indicates information that may be helpful.
	Shows a more detailed operation method or an item that describes additional information.

Expressions

Messages appear on the display panel as shown below.

- **A message displayed on the display panel (sample)**



---------- “U-1”

Contents

Introduction iii
DRYPIX Smart Manuals iv
Contents at a Glance v
How to Read This Manual vi

Chapter 1 For Safe Operation

1.1 Precautions Before Operating This Equipment 1-1
1.1.1 Precautions to be Observed When Using the Electric Medical Equipment 1-2
1.2 Safety 1-3
1.3 Electromagnetic Compatibility (EMC) 1-5
1.4 Handling and Storing Cautions for films for the Fuji Dry Imager 1-9
1.5 Locations of labels and signs 1-10
1.5.1 Locations 1-10
1.5.2 Safety and Other Signs 1-13
1.6 Equipment Installation Space 1-14
1.6.1 Installation Space 1-14
1.6.2 Space Required for Maintenance Work 1-14
1.7 Precautions on DRYPIX Smart Operations 1-15
1.7.1 Precautions for Use 1-15
1.7.2 Precautions When Outputting Film Prints 1-16

Chapter 2 Product Overview

2.1 Overview of the DRYPIX Smart 2-1
2.2 Features of the DRYPIX Smart 2-3
2.3 Units Names and the Functions 2-4
2.3.1 DRYPIX Smart Main Unit 2-4
2.3.2 Operation Panel 2-6

Chapter 3 Basic Operation

3.1 Starting / Shutting Down the Equipment 3-1
3.1.1 Starting the Equipment 3-1
3.1.2 Shutting Down the Equipment 3-2
3.2 Replacing the Film Pack 3-3
3.3 Replacing Film Tray 3-6

Chapter 4 Utility

4.1 Utility List 4-1
4.2 Starting Up the Utility 4-2
4.3 Quitting the Utility 4-3
4.4 Automatically Calibrating the Film Density 4-4
4.5 Outputting a QC Test Pattern 4-5
4.6 Resetting the Remaining Film Counter 4-6
4.7 Setting the Sensitive Material Correction Parameter 4-7
4.8 Setting the Date and Time 4-9

4.9 Outputting the SMPTE pattern 4-10
4.10 Outputting the Mammography QC Pattern 4-12

Chapter 5 Troubleshooting

5.1 In Case a Problem May Arise 5-1
5.2 The Equipment Cannot be Powered ON 5-2
5.2.1 The Equipment Does Not Start Up Normally 5-2
5.3 The Power to the Equipment Does Not Turn OFF 5-3
5.4 Error Code List 5-4
5.5 When a Film Jam Occurred 5-5
5.5.1 Film jamming when the error code “221” or “223” is displayed 5-6
5.5.2 Film jamming when the error code “226” or “227” is displayed 5-10
5.5.3 Film jamming when the error code “228” is displayed 5-11
5.5.4 Film jamming when the error code “230” is displayed 5-12
5.5.5 Film jamming when the error code “232” is displayed 5-13

Chapter 6 Care and Maintenance

6.1 Cleaning the Inside and Surroundings of the Equipment 6-1
6.1.1 Cleaning a film tray 6-2
6.1.2 Cleaning the Air Filter 6-3
6.1.3 Washing the Cleaning Roller (Optional) with Water 6-4
6.1.4 Image Verification 6-6
6.2 Users Checksheet (Care and Maintenance) 6-7
6.3 About Preventive Maintenance 6-8

Appendix A Specifications

A.1 Specifications A-1
A.2 External View and Weight A-2

Maintenance and Inspection

Chapter 1 For Safe Operation

1.1 Precautions Before Operating This Equipment

Before using this equipment, please read “Precautions Before Operating This Equipment” carefully so that you can operate it correctly.
Whenever you operate this equipment, be sure to observe those precautions. Failure to do so may cause you to subject to injuries or property damage to occur.

⚠WARNING

Do not connect any unspecified device.

This system is classified as a medical device under EC Directive 93/42/EEC.
This equipment has been designed on the assumption that the patient would not come into direct contact with it or for operation by appropriately trained operator.
The operator of this equipment must comprehend the contents shown in the labels or on the monitor.

Process waste correctly, as stipulated by local law or any regulations that apply.
When discarding the DRYPIX Smart that incorporates the lithium battery, be sure to contact a licensed waste disposal contractor because it cannot be disposed of as a general waste.
Not doing so may cause environmental pollution.

To install or relocate the equipment, request it to a designated service supplier.

Cautions on Network
After connecting this system to the network with other systems, confirm that the other systems are not affected. If they are affected, take countermeasures such as network separation.

Precautions on External Network Connection
When a setting of the network to which the equipment is connected has been changed, check that the change does not affect the system operation and take measures if necessary.
The setting change may include the following:
- Change of connection destination
- Addition of devices
- Removal of devices
- Update of devices
- Upgrade of devices

1.1.1 Precautions to be Observed When Using the Electric Medical Equipment

- Do not remodel the equipment.
- In an emergency, disconnect the power plug or the inlet.
  Ensure sufficient space around the power plug or the inlet.
- An additional MULTIPLE SOCKET-OUTLET or extension cord shall not be connected to this device.
- If the network is shut down due to a network device failure or any other cause, image data is no longer transferred to this device.

1.2 Safety

Before using the DRYPIX Smart, read this section thoroughly to ensure that you use the product properly.

Laser Handling Precautions

This equipment is a Class 1 Laser Product (IEC 60825-1/EN 60825-1).
This device contains an embedded laser with the following specification:

Class	: 3B
Medium	: Semiconductor laser
Wavelength	: 659nm (TYP)
Maximum output - Specification of Laser Diode	: 130mW (CW)
Maximum output - Fault Condition	: 400mW (CW)
Beam emittance	: 20°

⚠WARNING

To prevent the user from being exposed to laser beams, always observe the following precautions.

- **Never remove any covers other than the front cover, right-side cover and bottom right-side cover of the equipment.**
- **When opening covers for coping with film jams, strictly observe procedures set forth in this manual. Never perform any other procedures.**
- **Always contact Fujifilm service personnel immediately if a malfunction is suspected in the equipment.**

⚠CAUTIONS

Use of controls or adjustment or performance of procedures other than those specified herein may result in hazardous radiation exposure.

Electric Shock Warnings

⚠WARNING

The power supply to the DRYPIX Smart is AC 100/110/120V in North America and AC 200/220/230/240V in other countries.
To avoid electric shocks, users should always take the following precautions:

- **Only allow a trained operator to use the printer.**
- **Install the printer in a location where it will not be exposed to water.**
- **Check that the printer is securely earthed.**
- **Check that all of the cords and cables are completely and securely connected.**
- **When using the equipment within the environment where the patient may get into touch with it, optionally connect additional protective earth conductor.**
- **When using the equipment within the environment where the patient may get into touch with it, the user must not touch the equipment's exterior, such as covers and metal sections, and the patient at the same time. In addition, take care that the patient does not touch the equipment.**
- **To avoid risk of electric shock, this equipment must only be connected to a supply mains with protective earth.**

⚠ WARNING

Never remove the printer cover.
The printer contains high-voltage components that could cause an electric shock.

Ventilation precautions

⚠ CAUTIONS

Be sure to install this equipment in a well-ventilated environment.
Good ventilation must be ensured (at least 54m3/hour for one unit).

Precaution about High Temperature

⚠ CAUTIONS

Be careful with units where High-temperature Caution Labels are affixed as those units may be hot during operation.

Declaration of Conformity

This equipment is in conformance with the following standards:
North America
- UL 60601-1

Other countries
- IEC 60601-1/EN60601-1
- IEC 60601-1-2/IEC 60601-1-2

Classification

1) According to the type of protection against electric shock
   CLASS 1 EQUIPMENT
2) According to the degree of protection against electric shock
   NO APPLIED PART
3) Protection against harmful ingress of water or particulate matter
   IP00
4) According to the degree of safety of application in the presence of a flammable anesthetics mixture with air or with oxygen or nitrous oxide.
   Equipment not suitable for use in the presence of a flammable anesthetics mixture with air or with oxygen or nitrous oxide.
5) According to the mode of operation
   CONTINUOUS OPERATION

1.3 Electromagnetic Compatibility (EMC)

This equipment has been tested and found to comply with the limits for medical devices to the IEC60601-1-2+A1 (North America), IEC60601-1-2+A1 (EN60601-1-2+A1), IEC60601-1-2, Medical Device Directive 93/42/EEC (other than North America).
These limits are designed to provide reasonable protection against harmful interference in a typical medical installation.
This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to other devices in the vicinity.
However, there is no guarantee that interference will not occur in a particular installation.
If this equipment does cause harmful interference to other devices, which can be determined by tuning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the followingmeasures:

- Reorient or relocate the receiving device.
- Increase the separation between the equipment.
- Connect the equipment into an outlet on a circuit different from that to which the other device(s) are connected.
  Consult the manufacturer or field service technician for help.

Further information for IEC 60601-1-2 (EN60601-1-2)

- Medical electrical equipment needs special precautions regarding EMC and needs to be installed and put into service according to the EMC information provided in the accompanying documents.
- Portable and mobile RF communications equipment should not be used near this medical electrical equipment, as such communications equipment can affect this equipment and the images.
- Information regarding the cable affecting EMC is as follows.

Name	General Specification
Network Cable	Cat5e or more, UTP type and straight cable

- The use of accessories, transducers and cables other than those specified, with the exception of transducers and cables sold by FUJIFILM Corporation as replacement parts for internal components, may result in increased emissions or decreased immunity of the DRYPIX Smart.
- The DRYPIX Smart should not be used adjacent to or stacked with other equipment.
  If adjacent or stacked use is necessary, the DRYPIX Smart should be observed to verify normal operation in the configuration in which it will be used.

Guidance and manufacturer's declaration - electromagnetic emissions		
The DRYPIX Smart is intended for use in the electromagnetic environment specified below. The customer or the user of the DRYPIX Smart should assure that it is used in such an environment.		
Emissions test	**Compliance**	**Electromagnetic environment - guidance**
RF emissions CISPR 11	Group 1	The DRYPIX Smart uses RF energy only for its internal function. Therefore, its RF emissions are very low and are not likely to cause any interference in nearby electronic equipment.
RF emissions CISPR 11	Class A	The DRYPIX Smart is suitable for use in all establishments other than domestic and those directly connected to the public low-voltage power supply network that supplies buildings used for domestic purposes.
Harmonic emissions IEC61000-3-2	Class A	
Voltage fluctuations/ flicker emissions IEC61000-3-3	Complies	

Guidance and manufacturer's declaration - electromagnetic immunity			
The DRYPIX Smart is intended for use in the electromagnetic environment specified below. The customer or the user of the DRYPIX Smart should assure that it is used in such an environment.			
Immunity test	**IEC 60601 test level**	**Compliance level**	**Electromagnetic environment - guidance**
Electrostatic discharge (ESD) IEC61000-4-2	±6kV contact ±8kV air	±2kV contact ±4kV contact ±6kV contact ±2kV air ±4kV air ±8kV air	Floors should be wood, concrete or ceramic tile. If floors are covered with synthetic material, the relative humidity should be at least 30%.
Electrical fast transient/burst IEC61000-4-4	±2kV for power supply lines ±1kV for input/output lines	±2kV for power supply lines ±1kV for input/output lines	Mains power quality should be that of a typical commercial or hospital environment.
Surge IEC61000-4-5	±1kV differential mode ±2kV common mode	±1kV differential mode ±2kV common mode	Mains power quality should be that of a typical commercial or hospital environment.
Voltage dips, short interruptions and voltage variations on power supply input lines IEC61000-4-11	<5% U_T (>95% dip in U_T) for 0.5 cycle 40% U_T (60% dip in U_T) for 5 cycles 70% U_T (30% dip in U_T) for 25 cycles <5% U_T (>95% dip in U_T) for 5 s	<5% U_T (>95% dip in U_T) for 0.5 cycle 40% U_T (60% dip in U_T) for 5 cycles 70% U_T (30% dip in U_T) for 25 cycles <5% U_T (>95% dip in U_T) for 5 s	Mains power quality should be that of a typical commercial or hospital environment. If the user of the DRYPIX Smart requires continued operation during power mains interruptions, it is recommended that the DRYPIX Smart be powered from an uninterruptible power supply or a battery.
Power frequency (50/60Hz) magnetic fieldIEC61000-4-8	3 A/m	3 A/m	Power frequency magnetic fields should be at levels characteristic of a typical location in a typical commercial or hospital environment.
NOTE: U_T is the a.c. mains voltage prior to application of the test level.			

Guidance and manufacturer's declaration - electromagnetic immunity			
The DRYPIX Smart is intended for use in the electromagnetic environment specified below. The customer or the user of the DRYPIX Smart should assure that it is used in such an environment.			
Immunity test	**IEC 60601 test level**	**Compliance level**	**Electromagnetic environment - guidance**
Conducted RF IEC61000-4-6	3 Vrms 150 kHz to 80 MHz	3 Vrms	Portable and mobile RF communications equipment should be used no closer to any part of the DRYPIX Smart, including cables, than the recommended separation distance calculated from the equation applicable to the frequency of the transmitter.
Radiated RF IEC61000-4-3	3 V/m 80 MHz to 2.5 GHz	3 V/m	Recommended separation distance $d = 1.2\sqrt{P}$ $d = 1.2\sqrt{P}$ 80 MHz to 800 MHz $d = 2.3\sqrt{P}$ 800 MHz to 2.5 GHz where P is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer and d is the recommended separation distance in metres (m). Field strengths from fixed RF transmitters, as determined by an electromagnetic site survey,[a] should be less than the compliance level in each frequency range.[b] Interference may occur in the vicinity of equipment marked with the following symbol: 
NOTE 1: At 80 MHz and 800 MHz, the higher frequency range applies. NOTE 2: These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects and people.			
a Field strength from fixed transmitters, such as base stations for radio (cellular/cordless) telephones and land mobile radios, amateur radio, AM and FM radio broadcast and TV broadcast cannot be predicted theoretically with accuracy. To assess the electromagnetic environment due to fixed RF transmitters, an electromagnetic site survey should be considered. If the measured field strength in the location in which the DRYPIX Smart is used exceeds the applicable RF compliance, the DRYPIX Smart should be observed to verify normal operation. If abnormal performance is observed, additional measures may be necessary, such as reorienting or relocating the DRYPIX Smart. b Over the frequency range 150 kHz to 80 MHz, field strength should be less than 3 V/m.			

Recommended separation distances between Portable and mobile RF communications equipment and the DRYPIX Smart

The DRYPIX Smart is intended for use in the electromagnetic environment in which radiated RF disturbances are controlled.
The customer or the user of the DRYPIX Smart can help prevent electromagnetic interference by maintaining a minimum distance between portable and mobile RF communications equipment (transmitters) and the DRYPIX Smart as recommended below, according to the maximum output power of the communications equipment.

Rated maximum output power of transmitter W	**Separation distance according to frequency of transmitter** m		
	150 kHz to 80 MHz $d = 1.2\sqrt{P}$	80 MHz to 800 MHz $d = 1.2\sqrt{P}$	800 MHz to 2.5 GHz $d = 2.3\sqrt{P}$
0.01	0.12	0.12	0.23
0.1	0.38	0.38	0.73
1	1.2	1.2	2.3
10	3.8	3.8	7.3
100	12	12	23

For transmitters rated at a maximum output power not listed above, the recommended separation distance d in metres (m) can be estimated using the equation applicable to the frequency of the transmitter, where P is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer.

NOTE 1: At 80 MHz and 800 MHz, the separation distance for the higher frequency range applies.
NOTE 2: These guidelines may not apply in all situations.
Electromagnetic propagation is affected by absorption and reflection from structures, objects and people.

1.4 Handling and Storing Cautions for films for the Fuji Dry Imager

Please be aware of the following cautions for handling and storing of the film.

Precautions for storing or handling unused films

1 Use only the “Fuji Medical Dry Laser Image Film DI-HL” or “Fuji Medical Dry Laser Image Film DI-ML” that is compatible with this equipment. If a film other than DI-HL or DI-HM is used, a malfunction may be caused to the equipment.
2 Be sure to store unused films contained in the film pack in a cool, dry and dark place of low temperature and low humidity (temperature: 25°C or lower), avoiding radioactivity and reactive gases, same as for the conventional wet-type film.
3 The DI-HL films are light-sensitive. Do not open a film pack before it is loaded properly in the equipment.
4 Do not touch unused films with bare hands, otherwise adverse effects can appear on recorded images.
5 Do not take unused films out of the film pack that has once been loaded into the equipment and opened or add films to the film pack loaded, which will result in misoperation or failure of the equipment.
6 The film pack contains a protective sheet, in addition to the specified number of recording films. This protective sheet will remain in the film pack even after those recording films have been printed and it cannot be used for image recording. Because the protective sheet is a film, discard it appropriately together with other used films.

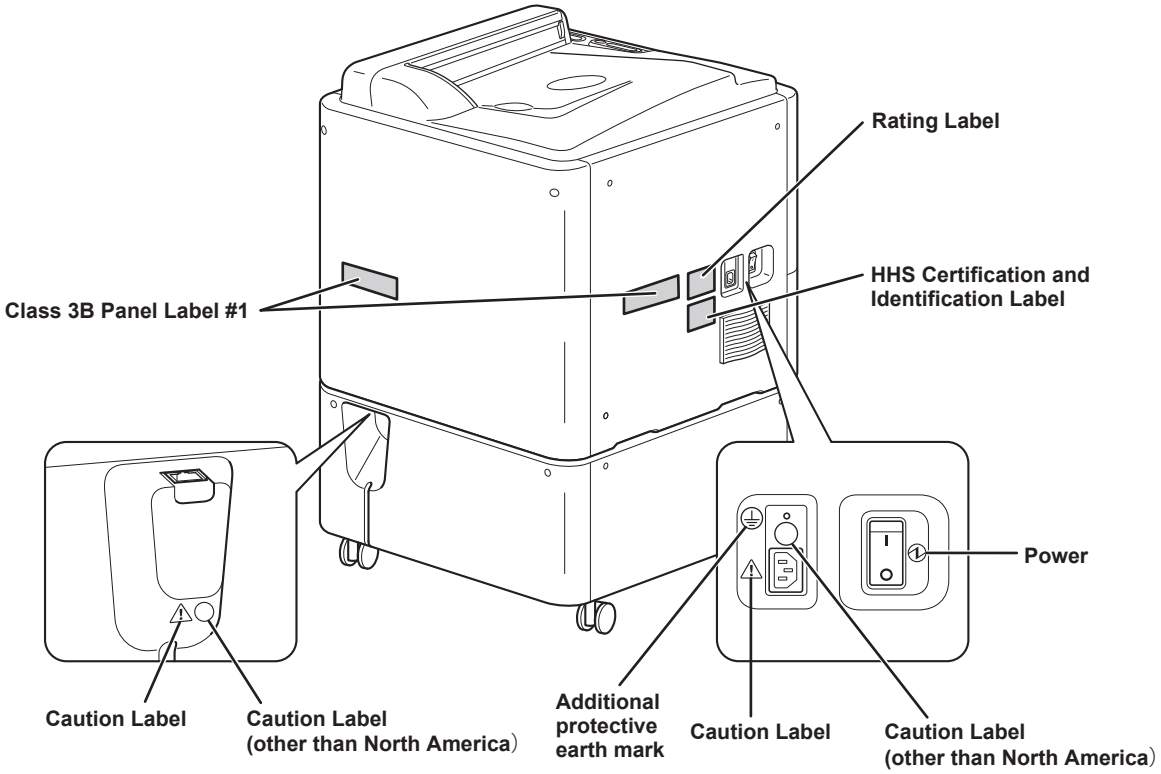
Precautions for storing or handling recorded films

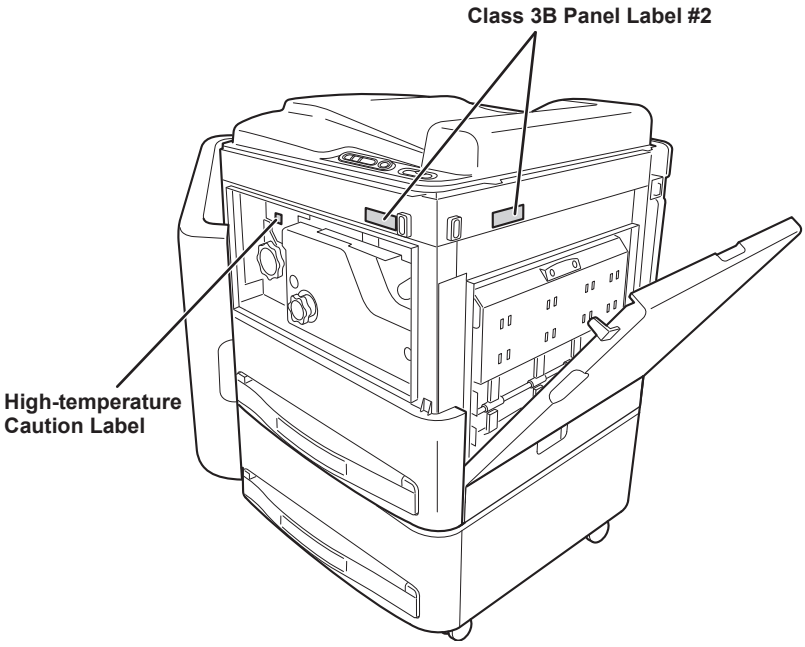
1 Store recorded films in a cool, dry and dark place of low temperature and low humidity. The higher the temperature and humidity, the more the density of recorded images will change sharply. Longterm storage at high temperature, high humidity and/or daylight conditions, such as in a car or room during summer, may cause discoloration. Using films in the slide projector or overhead projector will also cause discoloration.
2 For the long-term storage performance of recorded films, we assume based on the result of the acceleration test that it will be over 30 years at the storage temperature of 25ºC and over 25 years at 30ºC, until the portion on an image of density (D) = 1.2 at time of output to change 10% (ΔD=0.12).
3 After an image has been recorded, the film immediately after it was ejected from the machine is still in the process of image development and the room illumination or light emanating from the viewing boxwill cause slight changes in the optical density. Due to such optical effect, traces of overlapped films or transferred images can be visually recognized temporarily, which will disappear when those films are left under the normal light condition.
4 Note that lucid surfaces of recorded films can be lost or traces of contact with any chemicals that contain water, alcohol, developer, etc., and with other objects that contain a large amount of salt may appear on images, if they are handled under high-humidity environment or due to such undesirable contacts.
5 Do not store films with its image recording faces attached directly for preventing them from sticking to each other.

1.5 Locations of labels and signs

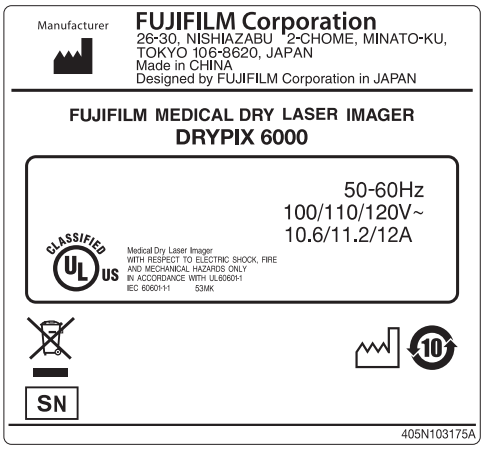
Locations of labels and signs affixed to the DRYPIX Smart, and the relevant safety signs are shown below.

1.5.1 Locations

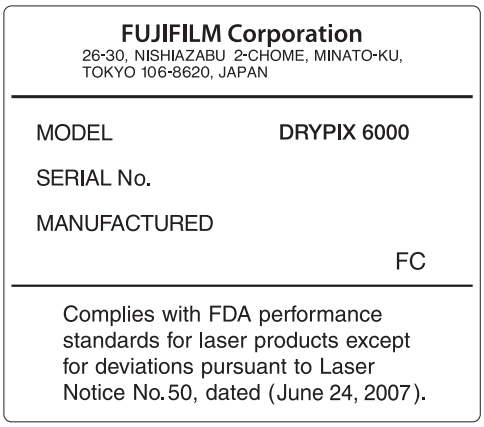




North America



Rating Label



HHS Certification and Identification Label



Class 3B Panel Label #1



Class 3B Panel Labels #2



Be aware that the temperature may be high in the area where the following label is affixed.

High-temperature Caution Label



Be sure to use the power cord our official dealer or local presentative provided. If an inappropriate cord is used, it may generate heat or fire.

Caution Label (Inlet)



Do not connect a telephone wire to the LAN connector. Only UTP-type straight LAN cables of Category 5e or higher are appropriate for connection to this connector.

Caution Label (LAN connector)

Other countries

Manufacturer

FUJIFILM Corporation
26-30, NISHIAZABU 2-CHOME, MINATO-KU,
TOKYO 106-8620, JAPAN
Made in CHINA
Designed by FUJIFILM Corporation in JAPAN

FUJIFILM MEDICAL DRY LASER IMAGER
DRYPIX 6000

50-60Hz
200/220/230/240V~
5.3/5.6/5.9/6A

SN

405N103176A

Rating Label

FUJIFILM Corporation
26-30, NISHIAZABU 2-CHOME, MINATO-KU,
TOKYO 106-8620, JAPAN

MODEL DRYPIX 6000

SERIAL No.

MANUFACTURED

FC

Complies with FDA performance standards for laser products except for deviations pursuant to Laser Notice No.50, dated (June 24, 2007).

HHS Certification and Identification Label

ja 注意 — ここを開くとクラス3Bのレーザ光が出ます。ビームの被ばくを避けてください。
en CAUTION – CLASS 3B LASER RADIATION WHEN OPEN. AVOID EXPOSURE TO THE BEAM.
zh 注意 — 打开这里时，会发出３B级的激光。请务必避开以免受到激光扫射。
es ATENCION – RADIACION LASER DE CLASE 3B EN CASO DE ABRIR AQUI. EVITAR LA EXPOSICION A LOS RAYOS.
ru ОПАСНО – ПРИ ОТКРЫТИИ В ДАННОМ МЕСТЕ ПРОИСХОДИТ ВЫБРОС ЛАЗЕРНОГО ИЗЛУЧЕНИЯ КЛАССА 3B. ИЗБЕГАЙТЕ ЛУЧЕВОГО ОБЛУЧЕНИЯ.
ar تنبيه - تعرض الشعاع الليزر من الفئة 3B عند فتح هيكل الجهاز. تجنب تعرضك للشعاع.

Class 3B Panel Label #1

CAUTION
CLASS 3B LASER RADIATION WHEN OPEN AND INTERLOCKS DEFEATED.
AVOID EXPOSURE TO THE BEAM.

Class 3B Panel Labels #2



Be aware that the temperature may be high in the area where the following label is affixed.

High-temperature Caution Label



Be sure to use the power cord our official dealer or local presentative provided. If an inappropriate cord is used, it may generate heat or fire.

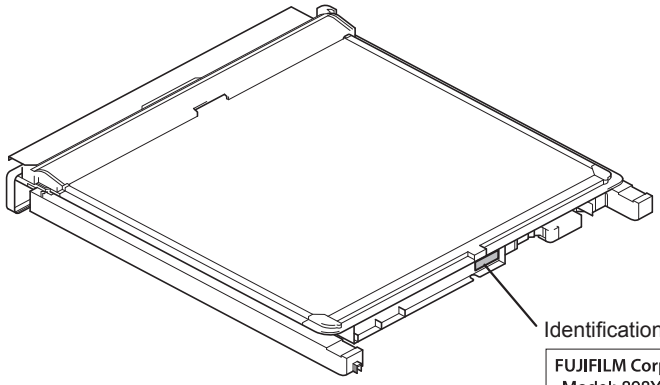
Caution Label (Inlet)



Do not connect a telephone wire to the LAN connector. Only UTP-type straight LAN cables of Category 5e or higher are appropriate for connection to this connector.

Caution Label (LAN connector)

Additional Option Tray



1.5.2 Safety and Other Signs

The following safety signs are used in the DRYPIX Smart labels or on its body.

Sign	Description
	This symbol indicates compliance of the equipment with Directive 93/42/EEC.
	Refer to operation manual
	Attention, consult ACCOMPANYING DOCUMENTS.
	High-temperature caution
	Power-OFF
	Power-ON
	Protective grounding (to the earth)
	Alternating current
	This symbol indicates that this product is not to be disposed of with your household waste, according to the WEEE Directive (2002/96/EC) and your national law. This product should be handed over to a designated collection point. Improper handling of this type of waste could have a possible negative impact on the environment and human health due to potentially hazardous substances that are generally associated with EEE. At the same time, your cooperation in the correct disposal of this product will contribute to the effective usage of natural resources. For more information about waste, please contact FUJIFILM dealers.
	Year of manufacture
	Environmentally Friendly Use Period (EFUP)
	Stand-by switch
	Enter button
	Utility button
	Serial number
	Manufacture

1.6 Equipment Installation Space

1.6.1 Installation Space

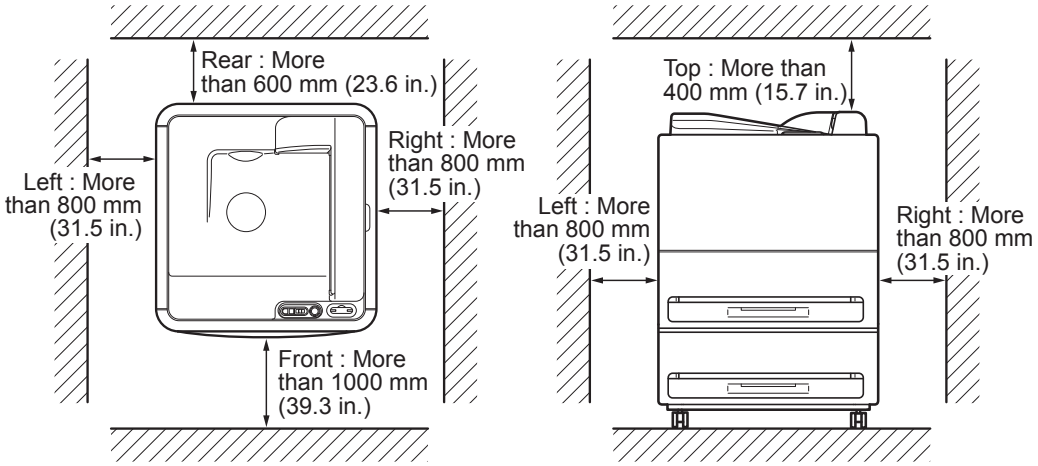
■ When Not Fixed with Fixtures

Front : More than 800 mm (31.5 in.)
Rear : More than 100 mm (3.9 in.)
Left : More than 50 mm (1.9 in.)
Right : More than 200 mm (7.87 in.)

■ When Fixed with Fixtures

Front : More than 800 mm (31.5 in.)
Rear : More than 100 mm (3.9 in.)
Left : More than 250 mm (9.8 in.)
Right : More than 250 mm (9.8 in.)

1.6.2 Space Required for Maintenance Work



1.7 Precautions on DRYPIX Smart Operations

Operations of image output on film are performed on the FCR PRIMA Console.
Therefore, no specific operation is required for printout on film for the DRYPIX Smart system.

1.7.1 Precautions for Use

⚠WARNING

This printer is not explosion-proof. Do not use flammable or explosive gases near this printer.

⚠CAUTIONS

When touching the keys of the control panel, be careful not to apply strong mechanical shock that would damage the keys.

⚠CAUTIONS

Be sure to check the patient information is printed on a film.

⚠CAUTIONS

If the film cutter breaks, replace it immediately. Take care to avoid injuring yourself on any sharp edges exposed by the breakage.
The film cutter is a consumable item. If it breaks, contact our official dealer and have it replaced.

⚠CAUTIONS

Sudden rises in the room temperature when the device is cold may cause dew condensation. Note that condensation in the device can lead to malfunctions.

⚠CAUTIONS

Be sure to cut the power supply during inspections.

⚠CAUTIONS

In the case of failure of the device, first cut the power supply and display a proper message (e.g. Out of order) in the device, then contact our official dealer.

⚠CAUTIONS

Do not lift and move the device. Contact our official dealer when you need to do.

⚠CAUTIONS

Do not get on the equipment or place any object on it. Otherwise, equipment breakage or human injury may result.

⚠ CAUTIONS

The weight of the equipment is approximately 104 kg (229.3 lb). Place the equipment in a location that can bear the weight.

⚠ CAUTIONS

Do not put weight on the film tray pulled out.

⚠ CAUTIONS

When using the shutter, be careful and hold it firmly with both hands to prevent it from dropping.

⚠ CAUTIONS

When locking a film tray or cover, be careful not to pinch your fingers or hands.

⚠ CAUTIONS

You may burn yourself if your bracelet or necklace touches high-temperature parts or units, such as inlet of the Thermal Development Unit.

1.7.2 Precautions When Outputting Film Prints

⚠ WARNING

If the power to the DRYPIX Smart is shut down unintentionally, image printouts may not be generated.
We, therefore, recommend that you do not erase images stored on the FCR/FDR Image Reader and other modalities, such as CT, MRI, and the like, before you make sure that images have been printed properly.
FUJIFILM Corporation and authorized dealers will not assume any responsibility for loss of images and others.

⚠ CAUTIONS

Do not forcibly remove a film in process of ejection.
Removing a film in process of ejection forcibly may damage it.

⚠ CAUTIONS

Do not open the front cover and right-side cover while images are being processed.
If a cover is opened while in processing, any image being processed at the moment will be disabled because correct print density cannot be obtained.
If a cover is opened while in processing, the equipment will stop operating for the safety reason. The alarm will then sound at the same time, displaying on the display panel a error code saying that a cover is open.

HINT

A maximum of 50 films can be stacked in the output film tray.
Remove the output films before the film tray becomes full.

HINT

Install this equipment in a well-ventilated place.

Chapter 2 Product Overview

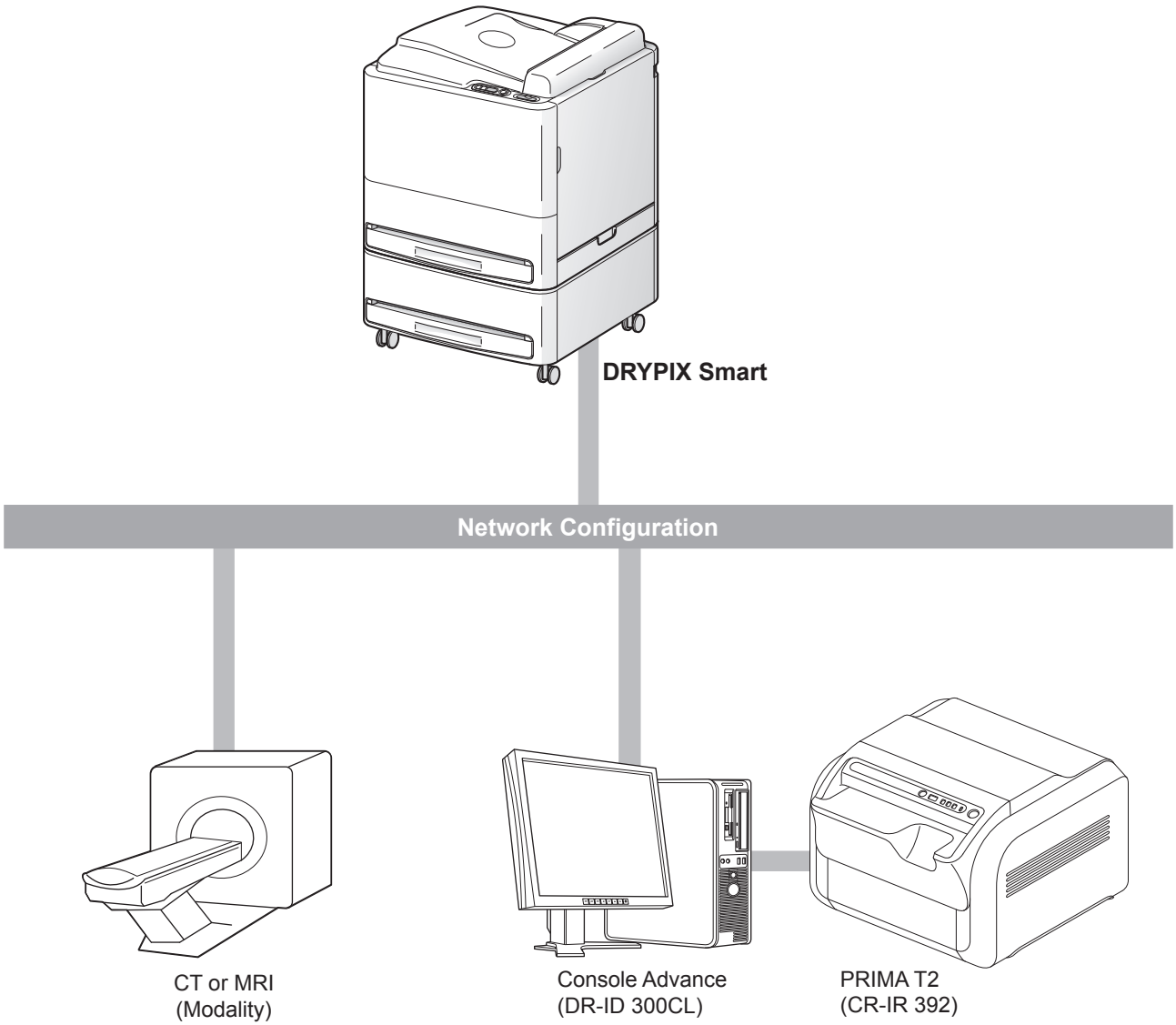
2.1 Overview of the DRYPIX Smart

■ Example of system configuration

The DRYPIX Smart is a device that prints digital image data, transferred via the DICOM network from the FCR/FDR Image Reader or DR, CT, MRI and other imaging modalities, onto a film.
It is also possible to print image information derived from imaging modalities or image processors not connected to the DICOM network.

Accessory equipment connected to the analog and digital interfaces must be certified according to the respective IEC standards (i.e. IEC 60950 for data processing equipment and IEC 60601-1 for medical equipment).
Furthermore all configurations shall comply with the system standard IEC 60601-1-1 and IEC 60601-1 Chapter 16. Everybody who connects additional equipment to the signal input part or signal output part configures a medical system, and is therefore, responsible that the system complies with the requirements of IEC 60601-1-1 and IEC 60601-1 Chapter 16. If in doubt, consult the technical services department or your local representative.
Use an IEEE 802.3 conformant HUB.

The figure below shows an example of a DRYPIX Smart connected to a network.



2.2 Features of the DRYPIX Smart

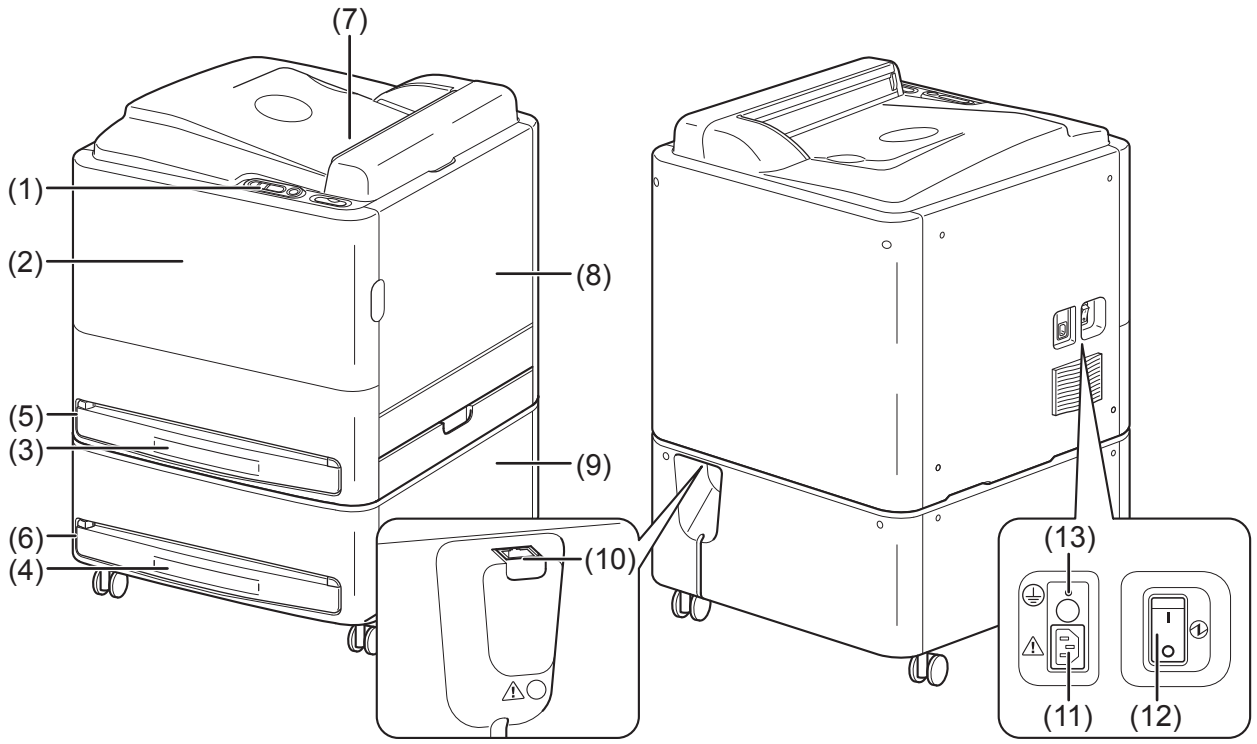
This section describes the main features of the Fuji Medical Dry Laser Imager DRYPIX Smart.

- This equipment adopts a system that records images with the laser and develops them by means of thermal processing, doing away with the need for management or replenishment of processing chemicals and disposal of waste solutions.
  This equipment thus uses an environment-friendly system.
- There is no need for darkrooms since the film packs can be replaced in bright locations.
- You can set up power saving mode to save power consumption of the equipment.
- When a film jams, you can correct it by opening covers as necessary.

2.3 Units Names and the Functions

This section names each of the DRYPIX Smart components and explains their functions.

2.3.1 DRYPIX Smart Main Unit



Name	Function
(1) Operation panel	Operation panel for the DRYPIX Smart.
(2) Front cover	Open this cover when a film jam occurs inside the equipment, and take appropriate actions to correct film jamming.
(3) Film tray 1 (4) Film tray 2	A film pack containing new films is loaded here. Note that basically this film tray is locked. For details on how to replace a film pack or pull out the film tray, see Pages 3-3 and 3-6 respectively.
(5) Slot 1 (6) Slot 2	A slot to insert the film tray.
(7) Output film tray	Ejected output films are stacked in this film tray. Up to 50 films can be stacked. If the stack exceeds 50 units, it may fall over.
(8) Right-side cover (9) Bottom right-side cover	Open this cover when a film jam occurs inside the equipment, and take appropriate actions to correct film jamming.
(10) LAN connector	Connected to the image network devices. 10base-T/100base-TX/1000base-T Interface
(11) Power inlet	Power inlet for this equipment.
(12) Main power switch	Basically leave this switch turned ON (**I**). ON (**I**) : Main Power ON OFF (**O**) : Main Power OFF
(13) Additional protective earth terminal	Additional protective earth terminal that the service engineer connects according to the system specifications.

⚠ CAUTIONS

Never open film tray or covers during processing.

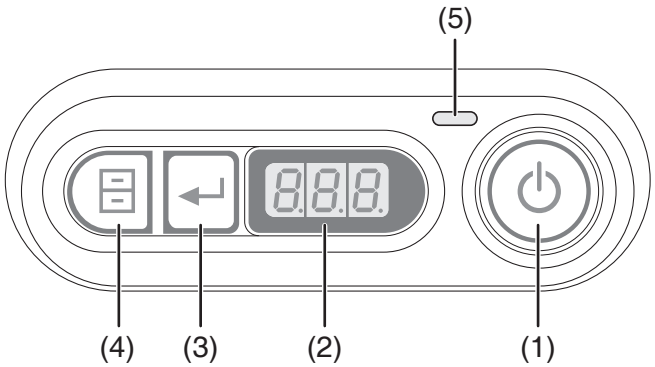
⚠ CAUTIONS

When closing covers, be careful not to pinch your finger in the aperture.

⚠ CAUTIONS

Do not connect telephone lines to LAN connector.
Only UTP-type straight LAN cables of Category 5E or higher are appropriate for connection to this connector.

2.3.2 Operation Panel



Name	Function
(1) Stand-by switch	When the stand-by switch is on, you can use this button to startup or shutdown the DRYPIX Smart.
(2) Display panel	Error code and the number of films remaining in each film tray are displayed here. Items shown below may also be displayed during printing.

Indication	Description
888	The equipment is starting up. (Between power-on and the initialization start)
ooo	The equipment is being initialized.
o x	The equipment is being initialized (x is the estimated time to boot up)
Pro	Printing is in progress.
oPn	The film tray is unlocked.
PUL	The film tray is locked.
SLP	The equipment is in the sleep mode.
oFL	The service utility is running. (Under work by a service engineer)
EP1/EP2/0-1/0-2	No more film
0~150	Display of the number of film sheets remaining * Normally tray 1. Press the “Enter button” on the operation panel to display Tray 2.
End	The equipment is shutting down.
U-0~U-7	User utility
SL1/SL2	Tray selection during the user utility
ESC	Cancel user utility
---	Tray retract indication

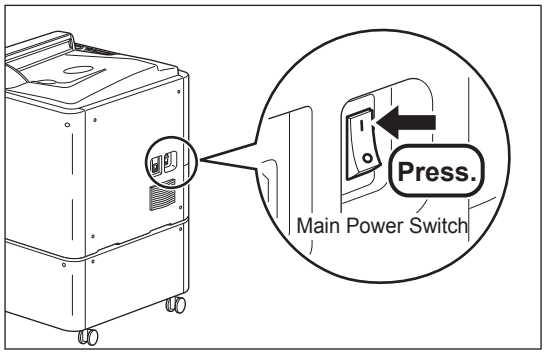
(3) Enter button	Unlocks the film tray when the film is empty or shutter is inserted. If pressed when the number of sheets remaining is displayed, the number of sheets remaining in film tray 2 will be displayed. It also will determine the operation to go to the next step in the Utility mode.
(4) Utility button	Pressing this button will enter Utility mode.
(5) Pilot lamp	Shows whether the main power is ON or OFF. Lit : The main power is on. Off : The main power is off.

Chapter 3 Basic Operation

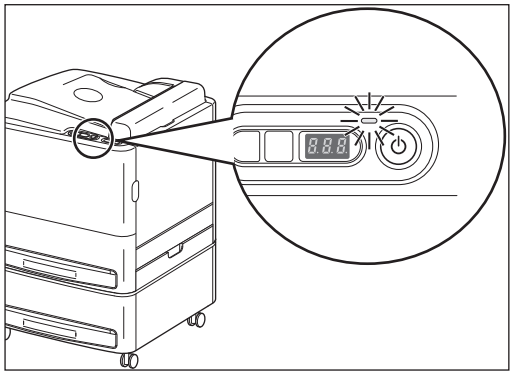
3.1 Starting / Shutting Down the Equipment

3.1.1 Starting the Equipment

1 Switch on the main power supply switch.



The pilot lamp turns on.



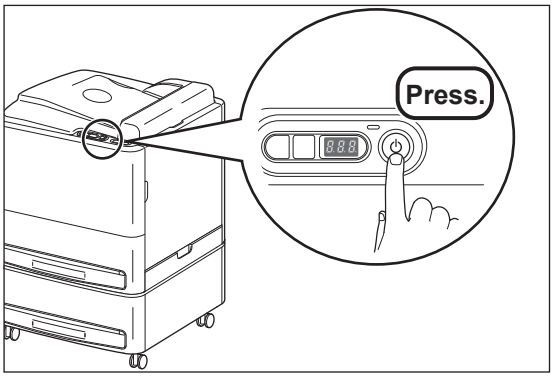
HINT

If the Main Power Switch is set to the “I” side, it is in the ON status.

2 Touch the stand-by Switch on the Operation Panel.

The DRYPIX Smart equipment starts running in about 10 to 15 minutes after the stand-by switch has been touched on the Operation Panel. See page 5-2 if the equipment does not start running properly.

Display panel will display “888” → “ooo” → “o x” → … → “o 1” while the printer is starting up.



3 When the equipment starts up, the number of remaining film sheets appears on the display panel.

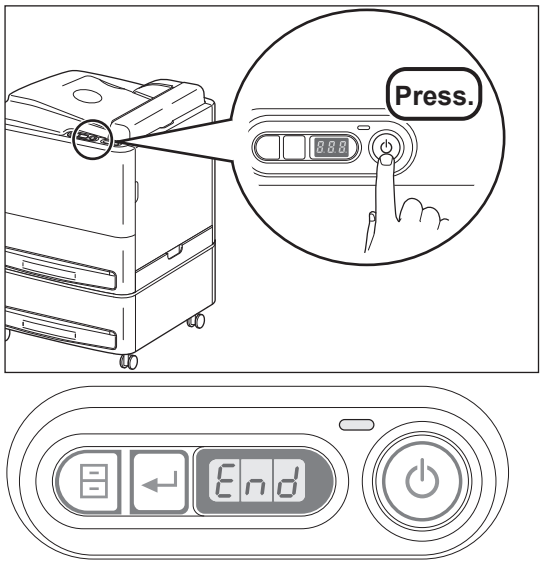
3.1.2 Shutting Down the Equipment

⚠ CAUTIONS

When “End” is displayed, the equipment is shutting down.
Do not turn off the main power switch when “End” remains displayed, otherwise the equipment may malfunction.
Do not turn off the main power switch until the “End” display disappears. The time period of the “End” display depends on the setting (approx. 10 to 30 sec.)
If “End” has been displayed for more than 10 minutes, an error may have occurred in the equipment. Turn the Main Power Switch off and contact Fujifilm service personnel.

See page 5-3 if the equipment does not shut down properly.

1 Hold down the stand-by Switch on the Operation Panel for a few seconds.



“End” appears on the Display panel.

⚠ CAUTIONS

If you want to reboot the equipment, wait for about 10 seconds or longer after the equipment has been shut down, and then touch the stand-by Switch.
This is to fully reset the system status inside the equipment.

3.2 Replacing the Film Pack

When it is printed with the film in the film tray set to “0”, “EP1” (Film Tray 1) or “EP2” (Film Tray 2) is displayed in the Display panel. Prepare Fuji Medical Dry Laser Image Film “DI-HL” or “DI-ML”, pull out the film tray, and replace the film pack.

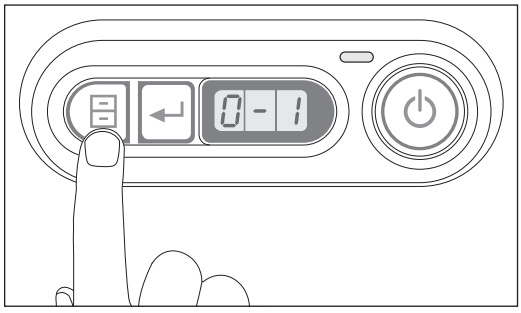
1 When the film in the film tray runs out during the film feeding, the film being fed is automatically output and “EP1” or “EP2” is displayed in the Display panel.

HINT

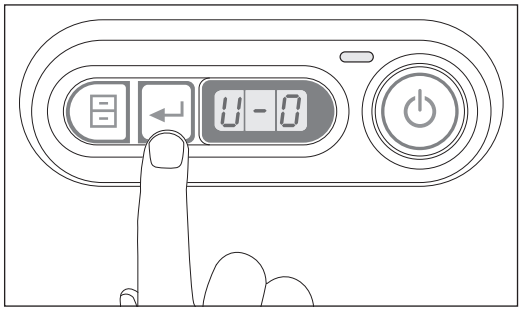
Use the procedure described below when the same film is loaded in both the film tray 1 and the film tray 2.

* The service setting is required in advance.

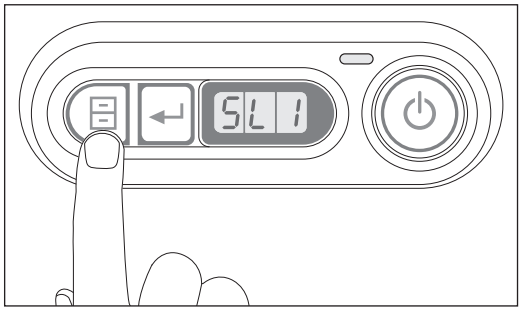
1. When the film in either of the film trays runs out, “0-1” appears for the film tray 1 and “0-2” for the film tray 2 on the Display panel, and an alarm sounds. Press the Utility button twice to stop the alarm and start up the utility.



2. Press the Enter button while “U-0” is displayed on the Display panel.

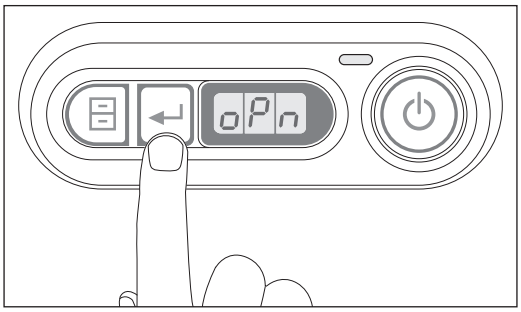


3. Press the Utility button to select the slot.



4. Select the slot and press Enter button to unlock the film tray lock. Go to Step **2**.

2 Press the Enter button and unlock the film tray lock.



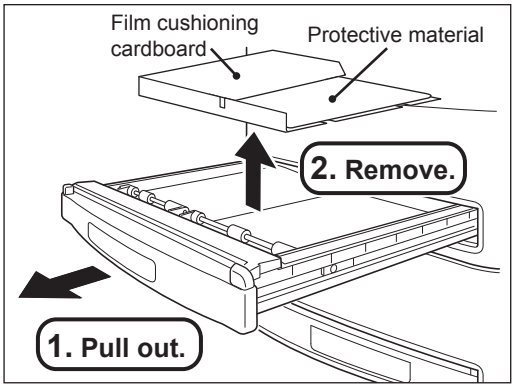
“oPn” appears on the Display panel.

CAUTIONS

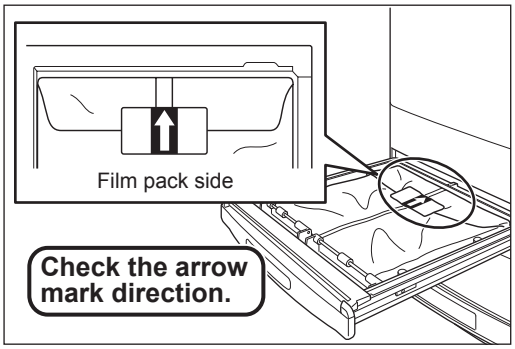
Be sure to pull the film tray out only after it has been unlocked. Otherwise, a system malfunction will result.

3 Pull the film tray out carefully (1.), and remove the protective material together with the film cushioning cardboard (2.).

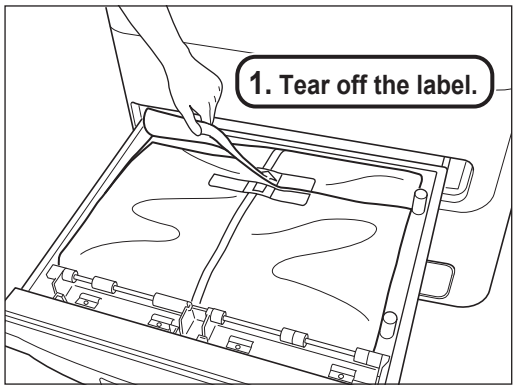
Discard the removed protective material and film cushioning cardboard.

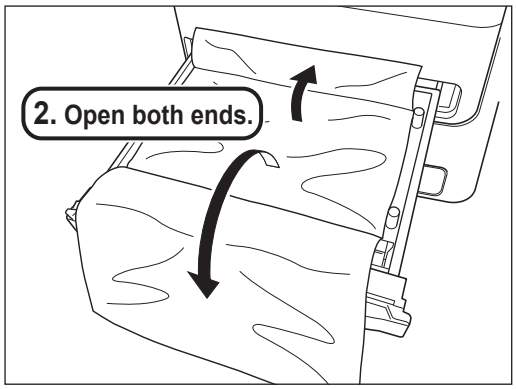


4 Make sure that the arrow mark on the new film pack points to the back side of the film tray.



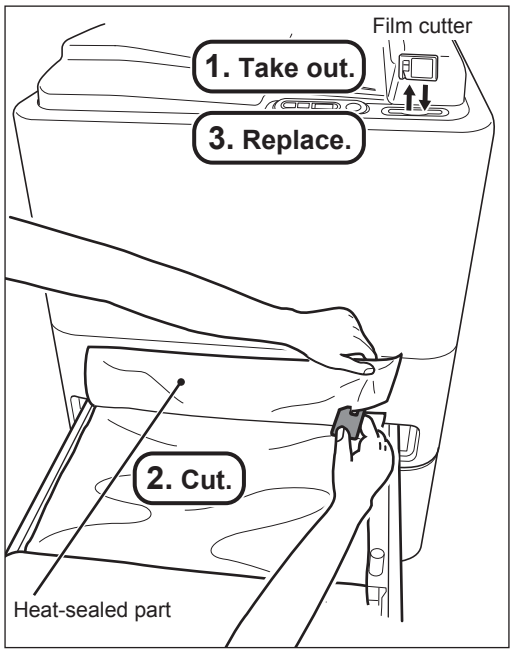
5 **Raise one end of the film pack to tear off the label (1.), and open both ends of the film pack (2.), as illustrated.**





6 **Use the film cutter stored on the right-hand side of the Operation Panel (1.) to cut the film pack along the film tray edge (2.). Replace then the film cutter in the position where it has been stored (3.).**

The film cutter is a consumable item. If the film cutter no longer cuts film properly or is damaged, contact our official dealer.



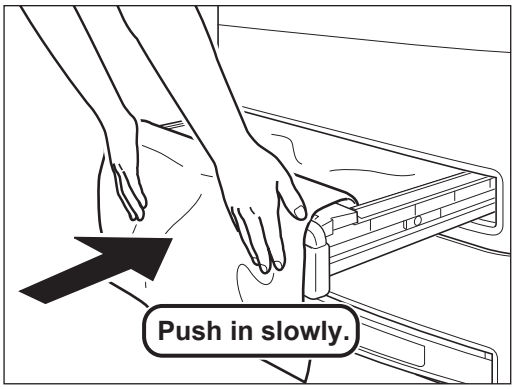
⚠ CAUTIONS

Make sure to cut off the heat-sealed part of the pack.

⚠ CAUTIONS

If the film cutter's blade protection is broken, the blade will be exposed posing then danger to the user.
Do not touch the unprotected blade.
Replace a dull film cutter hard to cut the film pack, as necessary.

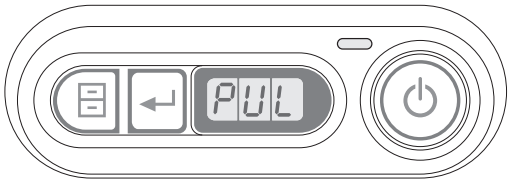
7 **Slowly push the film tray in until it locks firmly.**



⚠ CAUTIONS

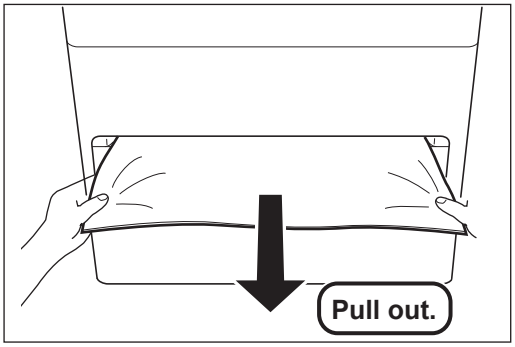
When closing the film tray, be careful not to pinch your fingers.

8 **Make sure that "PUL" is displayed on the Display panel.**



9 **Gently and slowly pull out straight toward you one film pack end that is outside the film tray.**

This completes the film supply procedure.



⚠ CAUTIONS

An attempt to pull the film pack out at a slant or roughly will cause it to break.

⚠ CAUTIONS

When “Waiting for size exchange” or “Film size error” is displayed on the status bar of “Queue list” in the image processor, check the display panel of the DRYPIX Smart and take the following measures.

- **If “oPn” is displayed on the display panel, no film is remaining in the film tray. Load new film.**
- **When the number of remaining film sheets appears on the display panel, replace the tray with one of a different film size or check the film size of the image processor.**

HINT

When a film pack is installed, the automatic film density calibration starts and when it finishes, the number of remaining film sheets will be displayed. The number of film sheets in the film tray will be changed.

HINT

When an error occurs during film feeding, first the film being fed is automatically output, and then the error code appears on the Display panel. When the stand-by Switch, Enter Button or Utility Button is pressed to stop the alarm and clear the error, “EP1” or “EP2” appears on the Display panel.

3.3 Replacing Film Tray

When you are pulling out a film-loaded tray, insert the shutter to avoid exposure to film.

CAUTIONS

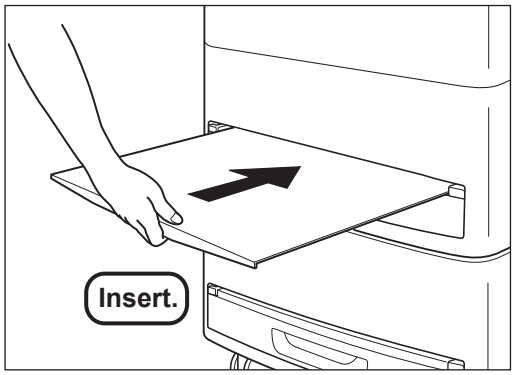
The table below shows the replaceable film trays.

The table below shows the replaceable film trays.
35 × 43 cm (14” × 17”), 35 × 35 cm (14” × 14”)
26 × 36 cm (10” × 14”), 25 × 30 cm (10” × 12”)
20 × 25 cm (8” × 10”) *

* 26 × 36 cm (10” × 14”), 25 × 30 cm (10” × 12”) and 20 × 25 cm (8” × 10”) films can be used together by the option setting.

1 Carefully place the shutter in the slit at the top of the film tray and fully insert it to the end.

Check that the Display panel is displaying the number of remaining film sheets when inserting the shutter.



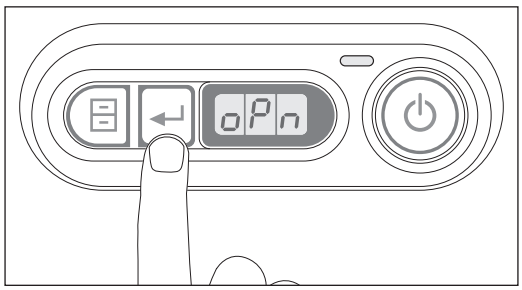
HINT

The shutter is stored in the rear-side of the DRYPIX Smart.

CAUTIONS

- **When inserting the shutter, be careful and hold it firmly with both hands to prevent it from dropping.**
- **When inserting or removing the shutter, take care that your hands are not caught between the parts.**

2 Press the Enter Button.



When the target film tray is unlocked, “oPn” appears on the Display panel.

CAUTIONS

The film tray 1 will be unlocked when the shutters are inserted into both the film tray 1 and the film tray 2.

CAUTIONS

The film tray can only be unlocked manually when the following conditions are met.

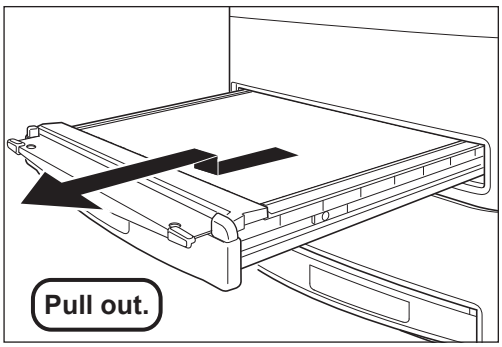
- **The device is in the idling state.**
- **There is one or more film remaining in the film tray.**
- **The shutter is inserted.**
- **The film tray is inserted.**

Under any of the following conditions, the film tray cannot be unlocked manually.

- **An image is being printed.**
- **The Utility mode is running.**
- **An error code is being displayed.**

The film tray cannot be unlocked manually during the starting, initializing or exiting the device.

3 Pull out the film tray together with the shutter until it stops, and then slightly lift the tray to completely pull it out.



CAUTIONS

When pulling out the film tray, be careful not to press the lock release button at the back.

CAUTIONS

Normally, the film tray is locked and cannot be pulled out as is. Do not forcibly pull out the tray. First press the Enter Button and make sure that “oPn” appears on the Display panel, and then pull out the tray.

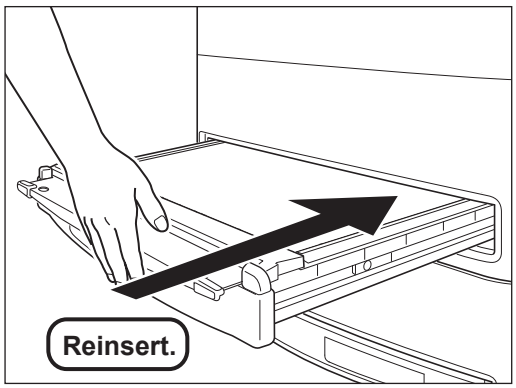
⚠ CAUTIONS

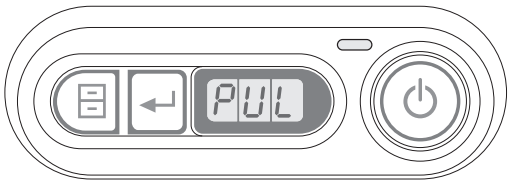
Do not remove the shutter while replacing.

⚠ CAUTIONS

When you remove the film tray, hold it firmly with your hands, and exercise care not to drop it on the floor.
The maximum weight of the film tray, including films, is approx. 8 kg (17.3 lb).

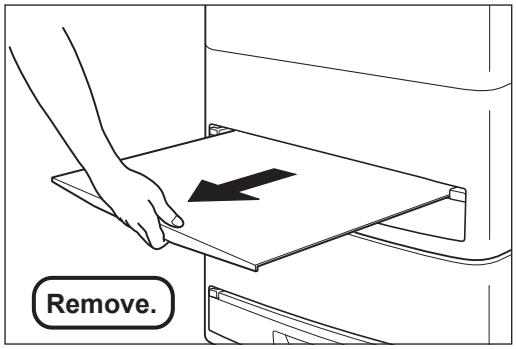
4 Insert the specified film tray together with the shutter.





"PUL" appears on the Display panel.

5 Push the film tray in firmly and make sure that the film tray has been locked correctly. Remove then the shutter.



HINT

Store the shutter in the rear-side of the DRYPIX Smart.

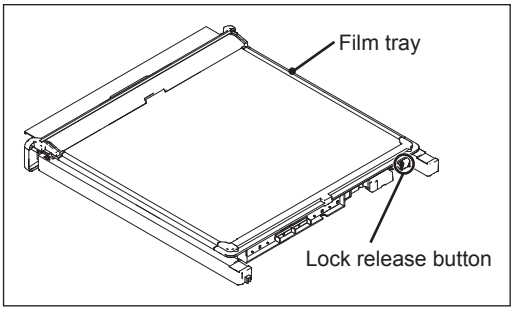
⚠ CAUTIONS

- **When removing the shutter, be careful and hold it firmly with both hands to prevent it from dropping.**
- **When inserting or removing the shutter, take care that your hands are not caught between the parts.**

The number of films remaining in the film tray when the tray is pulled out appears on the Display panel.

HINT

To remove the shutter when the film tray is not inserted, press the lock release button at the back.



⚠ CAUTIONS

If the shutter is removed when film remains in the film tray, the film may be exposed.

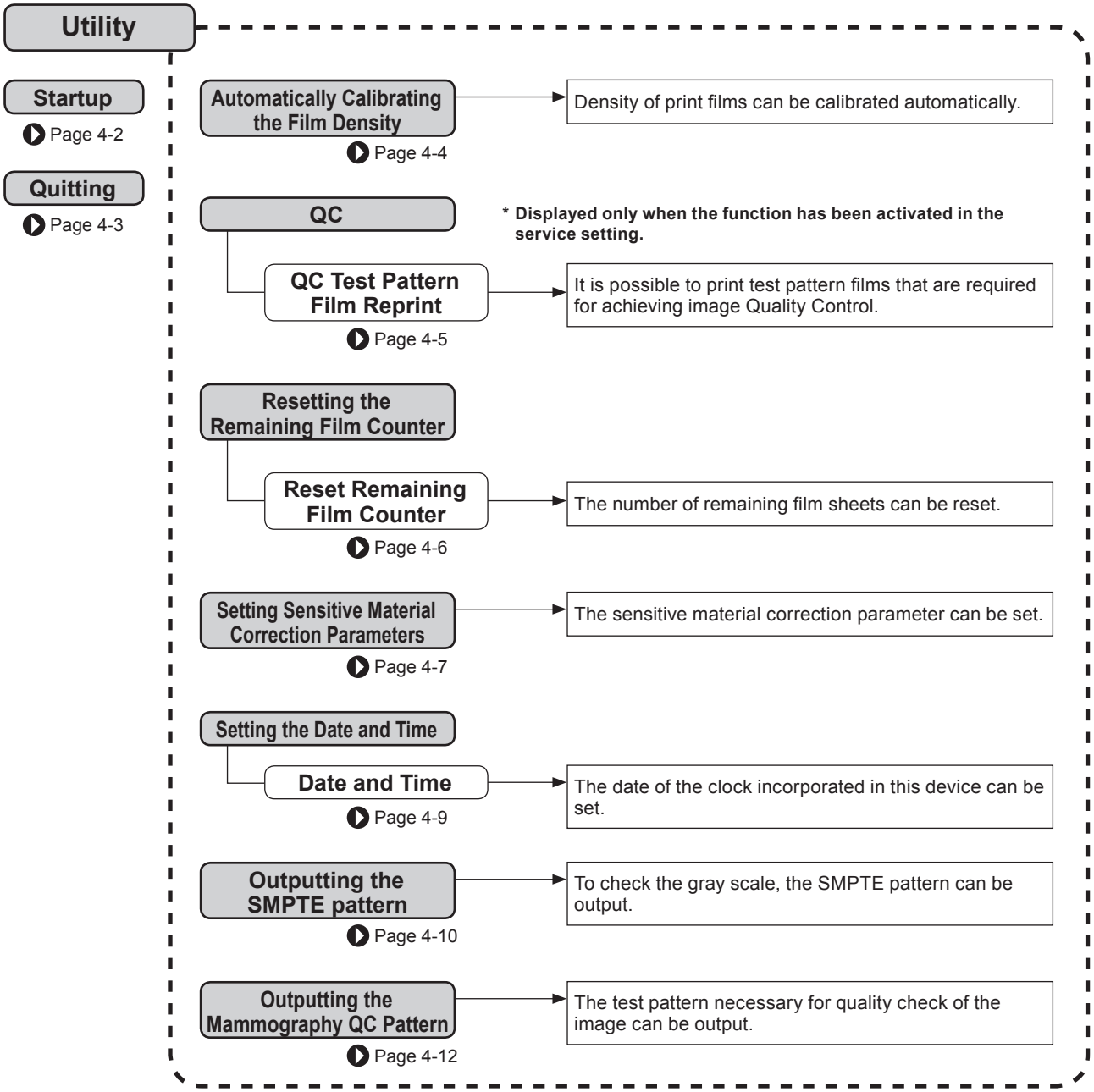
⚠ CAUTIONS

When an unregistered film tray is inserted or a different film tray is inserted, "219" appears on the Display panel and film tray is automatically unlocked.
Insert a different film tray.

Chapter 4 Utility

4.1 Utility List

Following are the functions available in the Utility mode. For details of the function operation procedures, see the respective pages for referencing shown in the flowchart below.



HINT

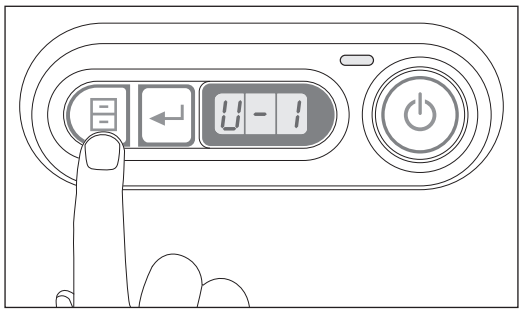
The table below shows the buttons available when the printer has switched to Utility mode.

Utility button		Switch to Utility mode. In the Utility mode, you can “display the next option”.
Set button		You can “confirm or execute the current option and proceed to the next step”.

4.2 Starting Up the Utility

Utility functions that the DRYPIX Smart system offers allow you to operate detailed processing or perform various settings.
Note that the Utility functions cannot be used while the equipment is in process of image printout.

1 **Press the Utility button.**



The display panel changes.

4.3 Quitting the Utility

Shown below is the procedure to quit the Utility mode.

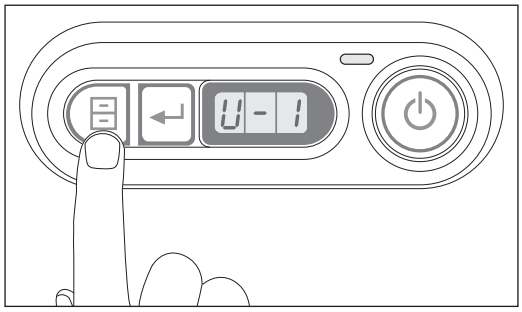
1 Display the number or remaining film sheets in either operation, and exit from the Utility.

- Press the Utility button few times.
- Press the Enter button when “ESC” is displayed.
- 10 seconds have elapsed without any operation

4.4 Automatically Calibrating the Film Density

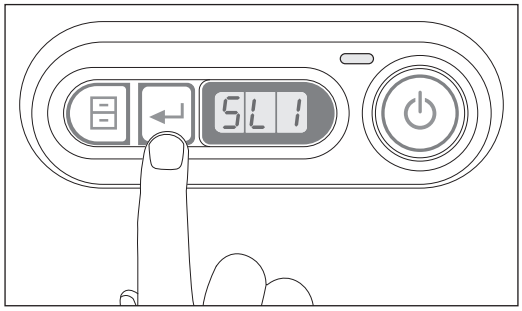
This function is used to calibrate the film density so that films are printed according to the density gradation curve value determined in the equipment.
Use this function when the film manufacturing number has been changed or the output density has changed.

1 **Press the Utility button.**



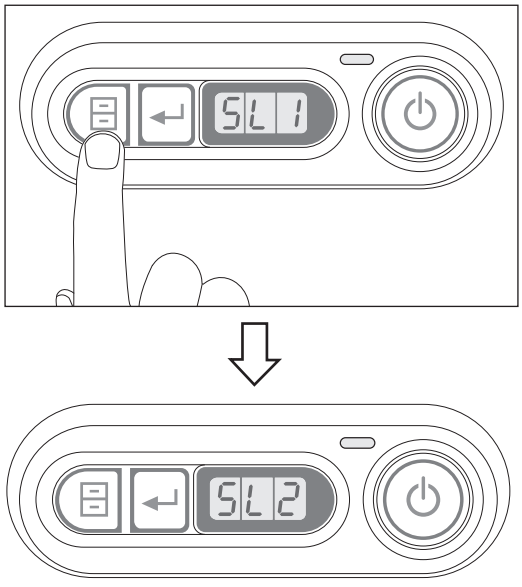
The Display panel changes and shows “U-1”.

2 **Press the Enter button while “U-1” is displayed.**

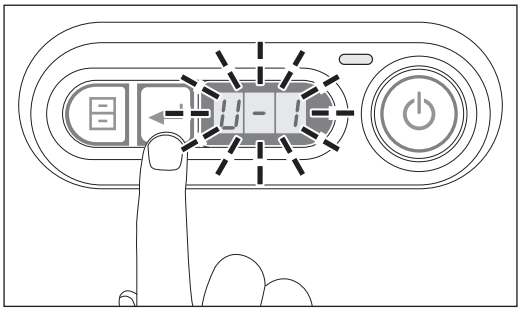


The slot 1 is displayed.

3 **Press the Utility button to select the slot.**

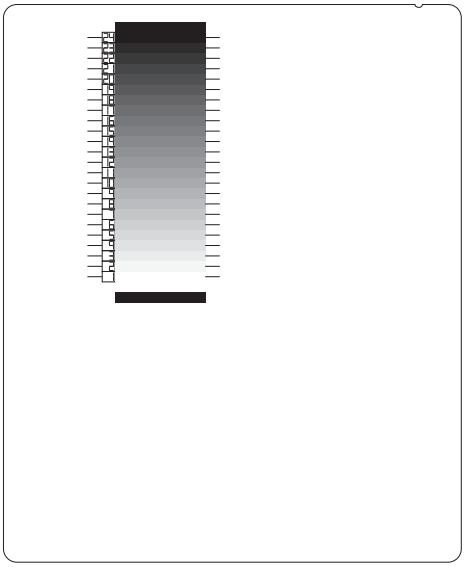


4 **Select the slot and press Enter button.**

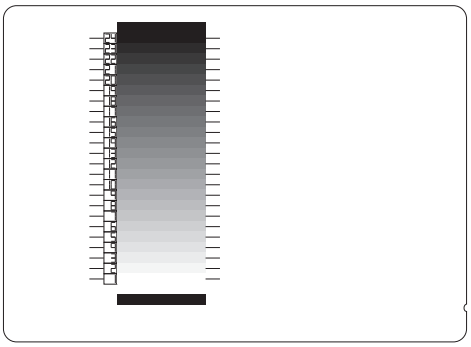


Auto gray level correction will be executed.
“U-1” in the Display panel blinks when auto gray level correction is executed.

Test pattern film (14”×17”)



Test pattern film (26×36cm)



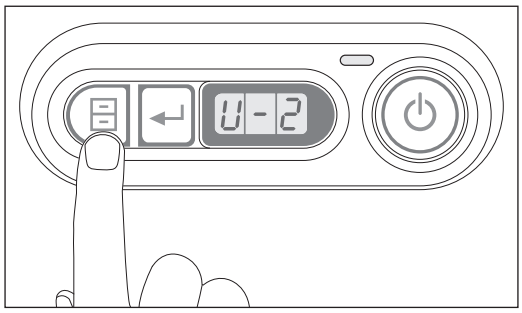
5 **When a test pattern is printed, the Utility mode automatically exits and the equipment becomes ready for printing again.**

If an error occurs, see the Chapter 5.

4.5 Outputting a QC Test Pattern

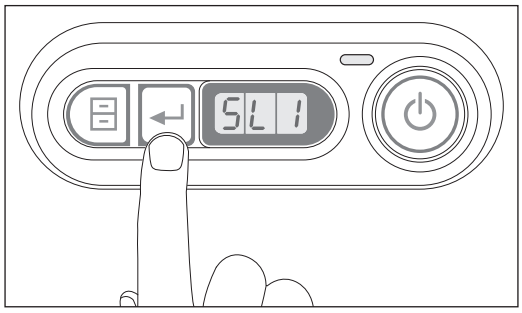
A QC test pattern can be output.
For details, see the “DRYPIX Smart Quality Control Function Reference Guide”.

1 Press the Utility Button twice.



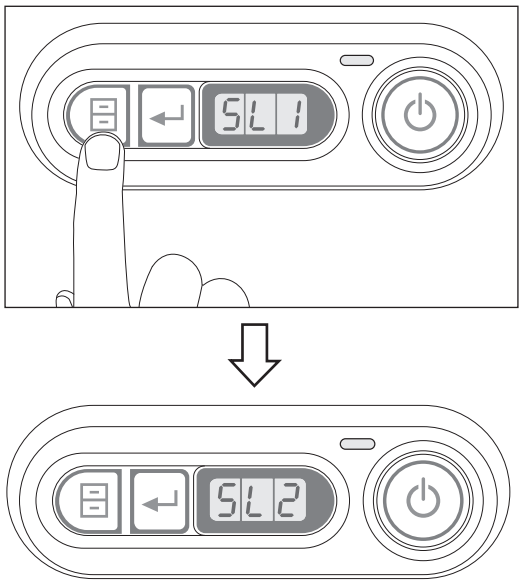
The Display panel changes and shows “U-2”.

2 Press the Enter button while “U-2” is displayed.

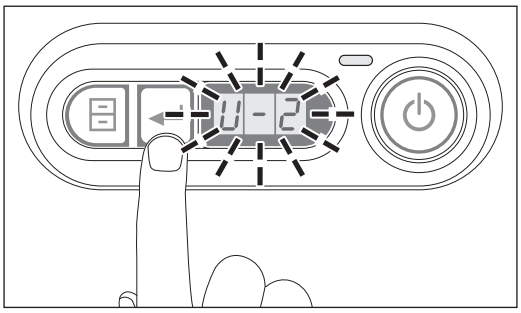


The slot 1 is displayed.

3 Press the Utility button to select the slot.



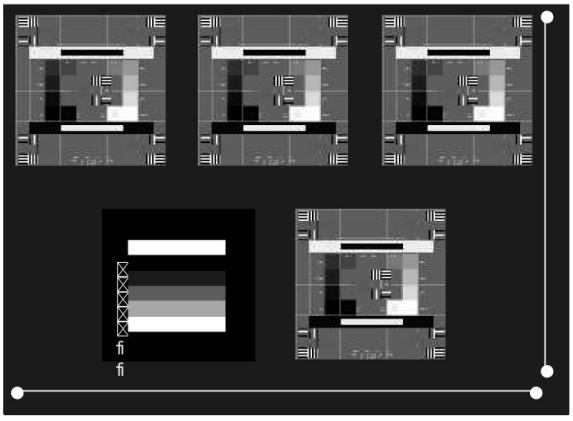
4 Select the slot and press Enter button.



QC output is executed.
“U-2” blinks when QC output is executed.

For detailson the QC function, see [DRYPIX Smart Quality Control Function Reference Guide].

5 When a QC test pattern is output, the Utility mode automatically exits and the equipment becomes ready for printing again.



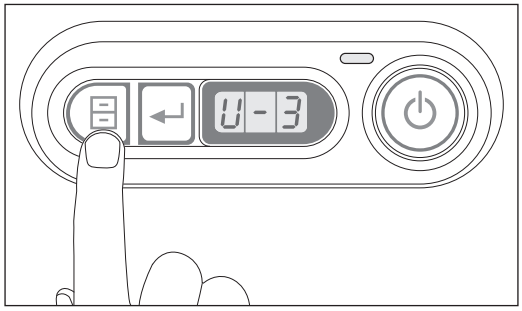
⚠ CAUTION

If an error occurs, see the Chapter 5.

4.6 Resetting the Remaining Film Counter

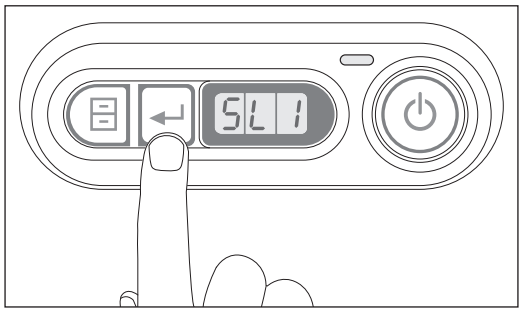
The number of remaining film sheets can be reset.

1 Press the Utility Button 3 times.



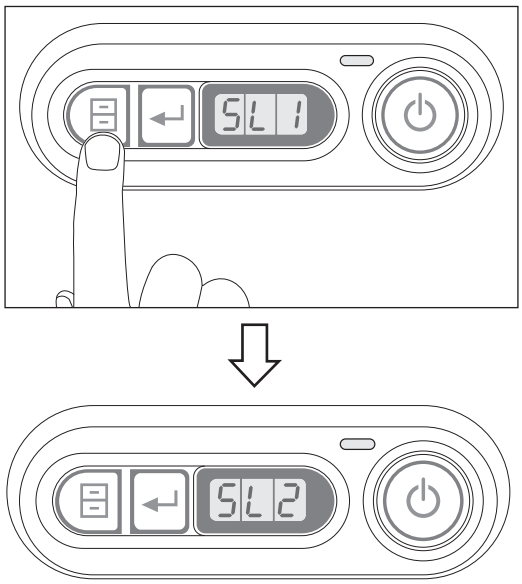
The Display panel changes and shows “U-3”.

2 Press the Enter button while “U-3” is displayed.

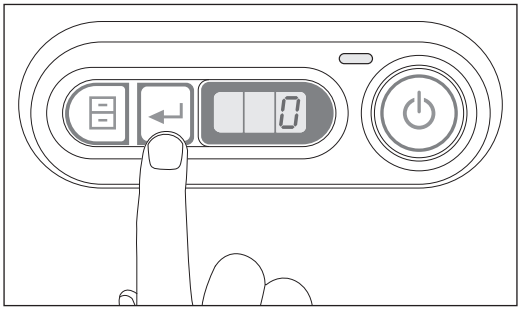


The slot 1 is displayed.

3 Press the Utility button to select the slot.



4 Select the slot and press Enter button.



The Remaining Film Counter is reset and “0” is displayed in the display panel.

⚠ CAUTIONS

- **The cumulative number of used films will not be reset.**
- **If the film tray is not inserted, an alarm sounds and the number of remaining film sheets cannot be reset.**

5 When the number of remaining film sheets is reset, the Utility mode automatically exits.

⚠ CAUTIONS

When an error occurs, the error code appears. When the error is cleared, the Utility mode automatically exits.

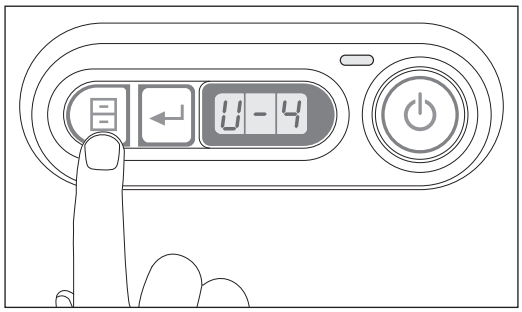
4.7 Setting the Sensitive Material Correction Parameter

The sensitive material correction parameter can be set.

⚠ CAUTIONS

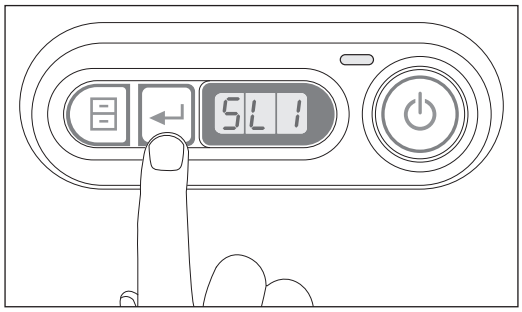
Use this function only when instructed to do so by Fujifilm service personnel. Do not use this without being instructed to do so.

1 Press the Utility Button 4 times.



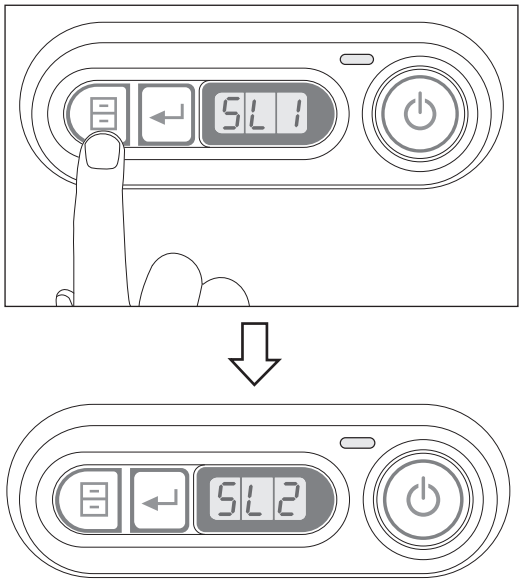
The Display panel changes and shows “U-4”.

2 Press the Enter button while “U-4” is displayed.

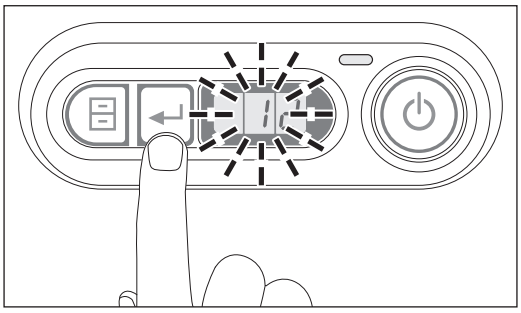


The slot 1 is displayed.

3 Press the Utility button to select the slot.



4 Select the slot and press Enter button.

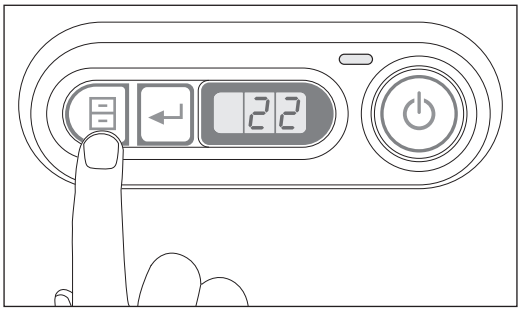


The tenth digit of the first half (consisting of two numbers) of the sensitive material correction parameter blinks.

⚠ CAUTIONS

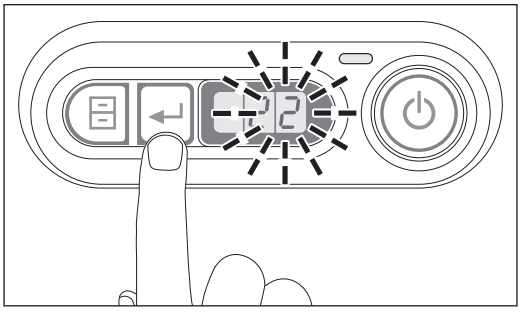
If the film tray is not inserted, an alarm sounds and the sensitive material correction parameter cannot be set.

5 Press the Utility button.



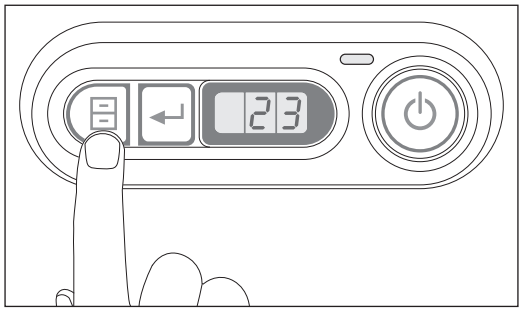
Every time the User Utility Button is pressed, the tenth digit of the first half (consisting of two numbers) of the sensitive material correction parameter increases in steps of 1.

6 Press the Enter Button.



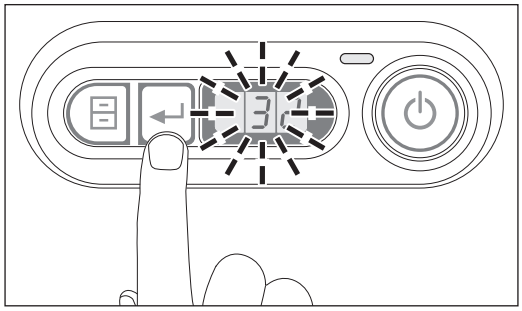
The first digit of the first half (consisting of two numbers) of the sensitive material correction parameter blinks.

7 **Press the Utility button.**



Every time the User Utility Button is pressed, the first digit of the first half (consisting of two numbers) of the sensitive material correction parameter increases in steps of 1.

8 **Press the Enter Button.**



The tenth digit of the last half (consisting of two numbers) of the sensitive material correction parameter (identified as the last two digits) blinks.

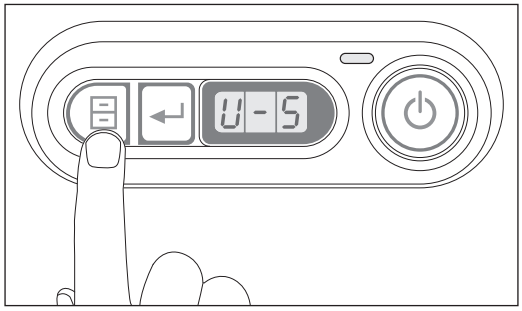
9 **Repeat Steps 3 to 5 to set the last 2 digits of the sensitive material correction parameter.**

10 **Pressing the Enter Button determines the sensitive material correction parameter, and the Utility mode automatically exits and equipment becomes ready for printing again.**

4.8 Setting the Date and Time

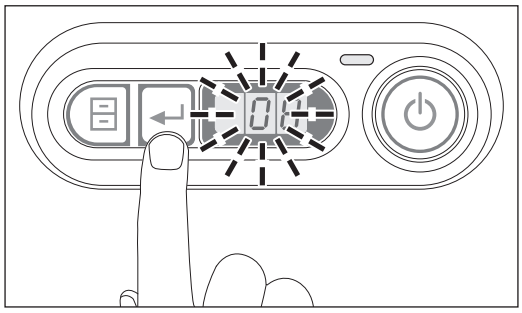
The date of the clock incorporated in this device can be set.

1 **Press the Utility Button 5 times.**



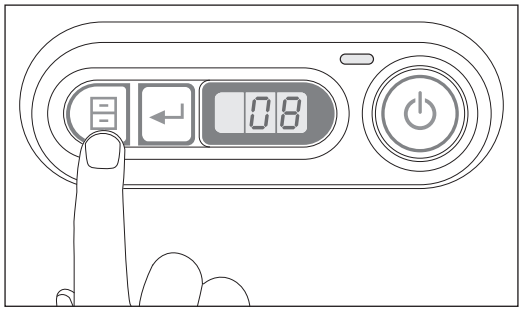
The Display panel changes and shows “U-5”.

2 **Press the Enter Button.**



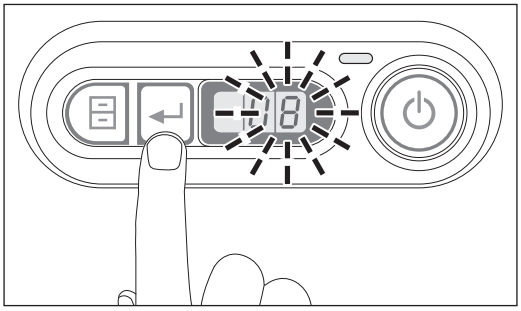
The tenth digit of “Year” blinks.

3 **Press the Utility button.**



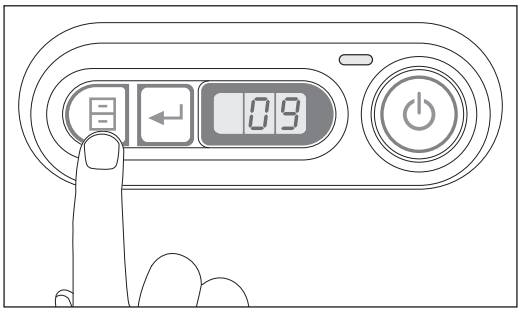
Every time the User Utility Button is pressed, the tenth digit of “Year” increases in steps of 1.

4 **Press the Enter Button.**



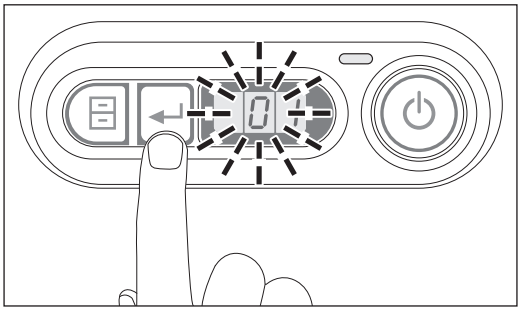
The first digit of “Year” blinks.

5 **Press the Utility button.**



Every time the User Utility Button is pressed, the tenth digit of “Year” increases in steps of 1.

6 **Press the Enter Button.**



The tenth digit of “Month” blinks.

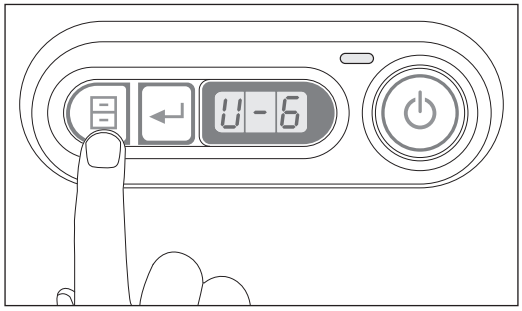
7 **Repeat Steps 3 to 5 four times to set parameters of month, day, hour and minute.**

8 **Pressing the Enter Button determines the date and time, and the Utility mode automatically exits and equipment becomes ready for printing again.**

4.9 Outputting the SMPTE pattern

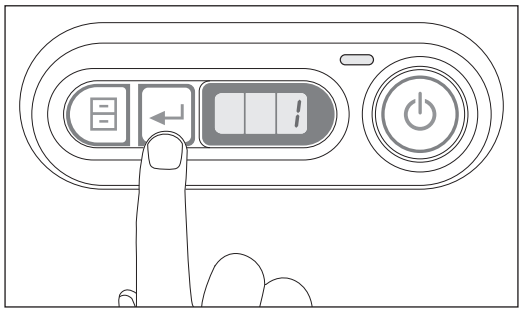
To check the gray scale, the SMPTE pattern can be output.

1 **Press the Utility Button 6 times.**



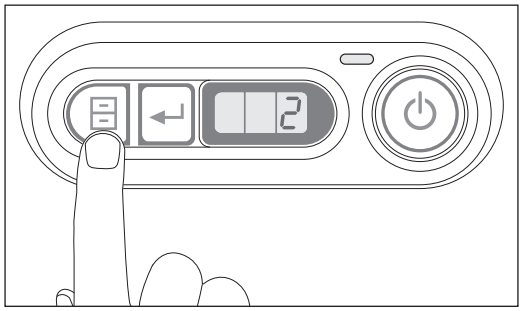
The Display panel changes and shows “U-6”.

2 **Press the Enter button.**



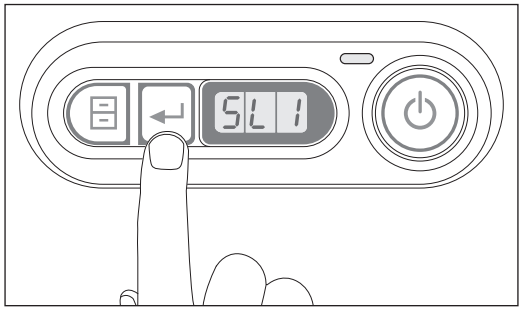
Display panel will switch to setting of the SMPTE pattern.

3 **Press the Utility button to set the SMPTE pattern.**



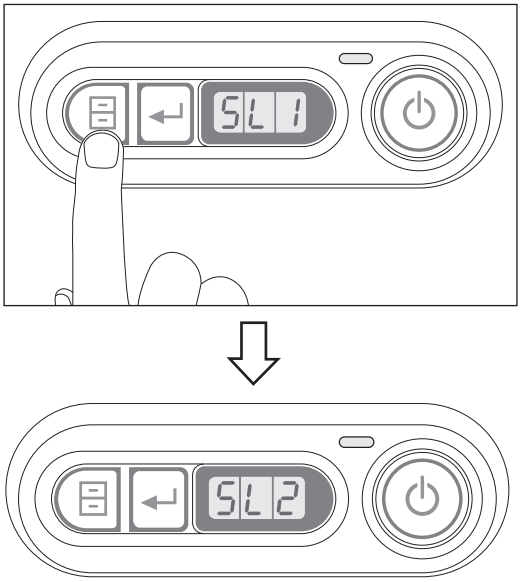
Each press of the Utility button switches the setting for the SMPTE pattern (1 - 5).

4 **Press the Enter button.**

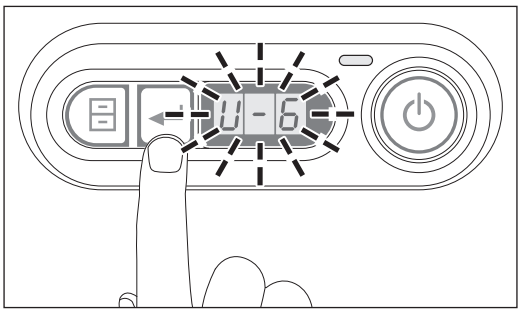


The slot 1 is displayed.

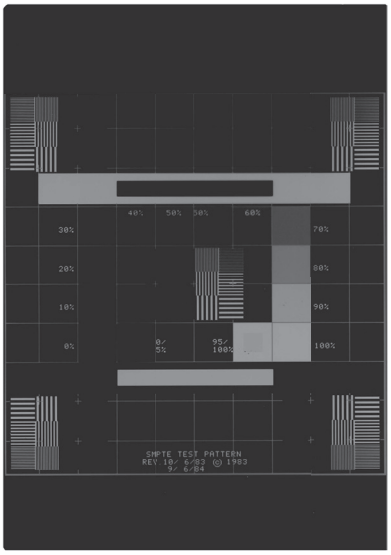
5 **Press the Utility button to select the slot.**



6 **Select the slot and press Enter button.**



“U-6” blinks while the output of SMPTE pattern is output.



⚠ CAUTIONS

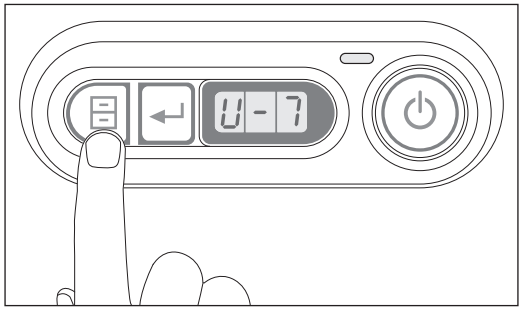
Output patterns differ depending on the default settings of the equipment.
To change the output patterns, consult FUJIFILM service personnel.

7 **When the SMPTE pattern is output, and the Utility mode automatically exits and equipment becomes ready for printing again.**

4.10 Outputting the Mammography QC Pattern

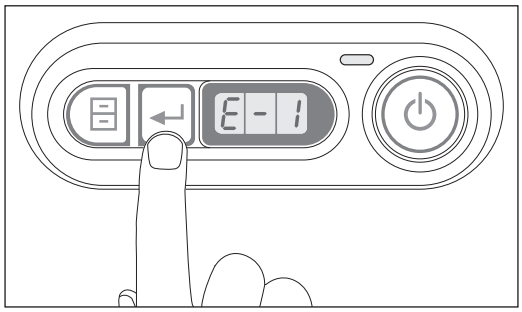
The mammography QC pattern can be output.
Refer to “DRYPIX Smart Quality Control for Mammography Reference Guide” for more information.

1 **Press the Utility button 7 times.**



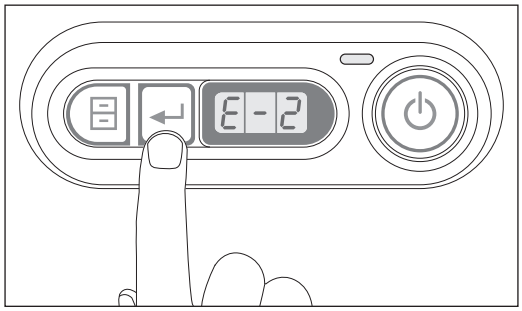
The display panel changes and shows “U-7”.

2 **Press the Enter button.**



Display panel will switch to setting of the mammography QC pattern.

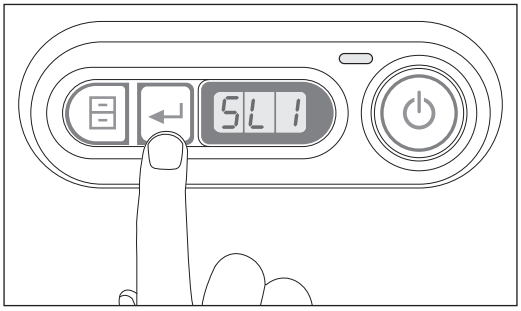
3 **Press the Utility button to set the mammography QC pattern.**



Each press of the Utility button switches the setting for the mammography QC pattern.
“E-1”→“E-2”→…→“E-7”→“n-1”→“n-2”→…→“n-5”→“ESC”

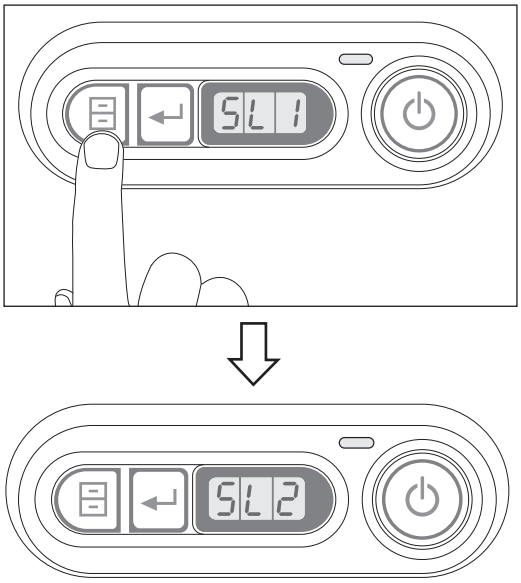
Panel display	Image name	Panel display	Image name
E-1	TG18-QC	n-1	TG18-QC
E-2	TG18-PQC	n-2	TG18-PQC
E-3	TG18-PQC (HIGH)	n-3	17-Steps
E-4	TG18-UN80	n-4	Grid
E-5	TG18-UN10	n-5	Special Resolution
E-6	TG18-UNL80		
E-7	TG18-UNL10		

4 **Press the Enter button.**



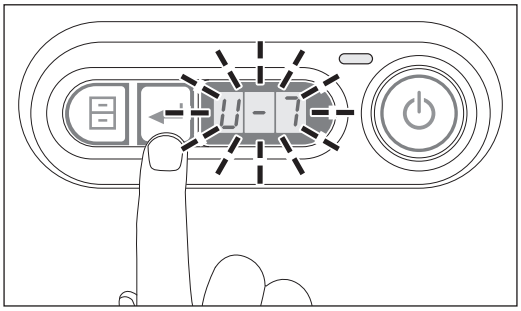
The slot 1 is displayed.

5 **Press the Utility button to select the slot.**



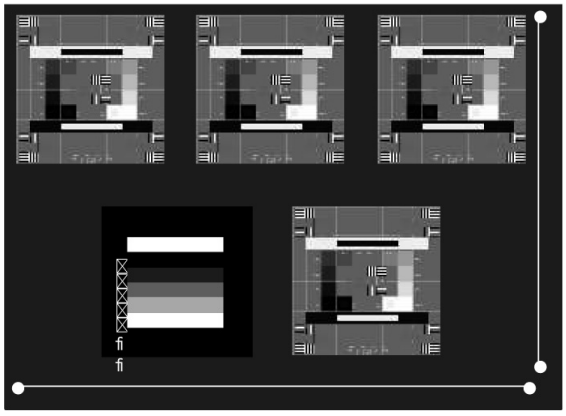
Each press of the Utility button switches the setting for the mammography QC pattern.
(“E-1”→“E-2”→…→“n-1”→“n-2”→…→“ESC”)

6 **Select the slot and press Enter button.**



“U-7” blinks while the output of mammography QC pattern is output.

7 **If outputting a mammography QC pattern, the Utilities are terminated automatically, and the device returns to a status where you can print.**

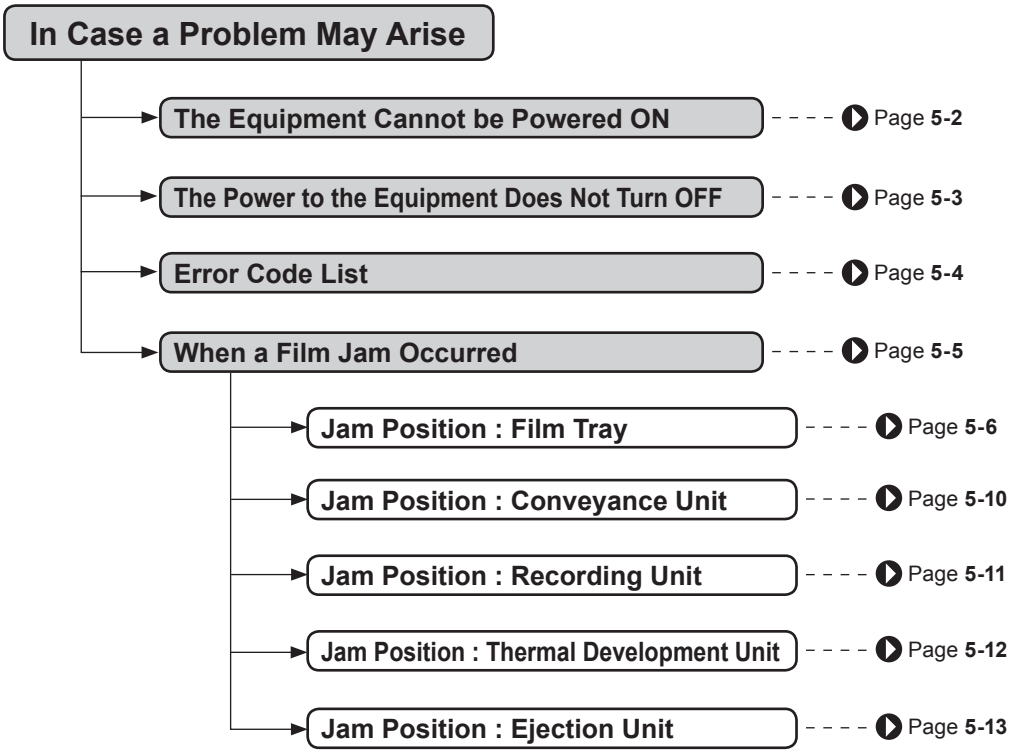


When an error occurs, refer to Chapter 5.

Chapter 5 Troubleshooting

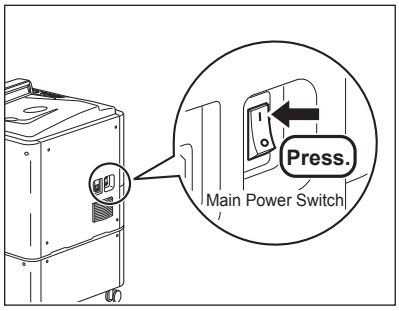
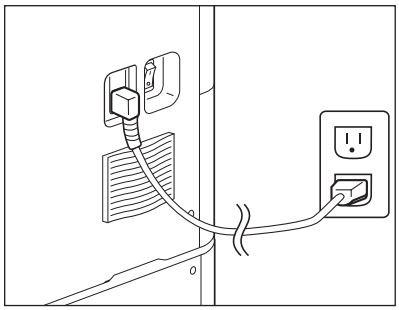
5.1 In Case a Problem May Arise

When the equipment does not operate normally, see the relevant page for you to take necessary actions appropriately, according to the symptom shown below.
If the problem cannot be solved, contact Fujifilm service personnel.



5.2 The Equipment Cannot be Powered ON

The equipment cannot be powered ON even at the press of the stand-by Switch.

The Main Power Switch has been turned OFF.	☞ Set the Main Power Switch to the "|" side to turn it ON. When power is supplied, the pilot lamp lights up.	
The power cord has been disconnected.	☞ Turn the Main Power Switch OFF and then plug the power cord firmly in the outlet. Turn the Main Power Switch on and then press the Stand-by switch. When power is supplied, the pilot lamp lights up.	

5.2.1 The Equipment Does Not Start Up Normally

Even though the power has been turned ON, the equipment does not start up normally.

Nothing appears on the Display panel

Contact FujiFilm service personnel.

An error code appears while the device is in the process of startup

An equipment cover is open.	☞	When the cover is open or mistakenly opened during the startup, an alarm sounds and the error code "201" appears on the Display panel. Press the stand-by Switch, Enter Button or Utility Button to stop the alarm, and then close the cover. When the cover is closed, the device reboots.
The film tray is pulled out.	☞	When the film tray is pulled out, an alarm sounds and the error code "203" or "207" appears on the Display panel. Press the stand-by Switch, Enter Button or Utility Button to stop the alarm. Contact Fujifilm service personnel as the film tray may be defective.
The shutter is inserted in a film tray.	☞	When the shutter is inserted, an alarm sounds and the error code "204", "208", "205" or "209" appears on the Display panel. Press the stand-by Switch, Enter Button or Utility Button to stop the alarm and remove the shutter on the film tray. Remove the shutter to reboot the device.

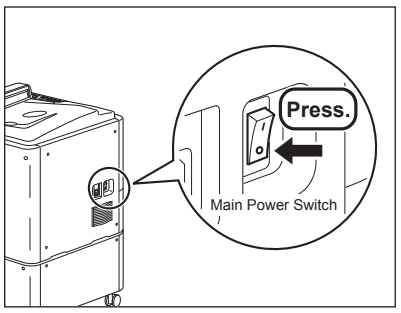
5.3 The Power to the Equipment Does Not Turn OFF

The device does not exit even if the stand-by Switch is held down for a few seconds.

An error code does not appear

An unprocessed film remains in the equipment.	☞	After the film output is finished, hold down the stand-by switch for a few seconds and then turn the power off.

“End” remains displayed for more than 10 minutes.

10 minutes or more has passed since “End” appears on the display panel.	☞	Turn OFF (press to the O side) the Main Power Switch at the bottom right-side corner on the side of the equipment.	

⚠ CAUTIONS

Normally “End” is displayed for 10 to 30 seconds after pressing the Stand-by switch. Do not turn off the Main Power Switch during this period, otherwise the equipment may malfunction.

5.4 Error Code List

When an error occurs on the device, an alarm sounds and the error code appears on the Display panel.
Press the stand-by Switch, Enter Button or Utility Button to stop the alarm.
The error codes appearing on the Display panel are listed below.

Error code	Actions
201	Check that the right side cover is closed, and close the front cover.
203/207	Load the tray. (Slot 1/Slot 2)
204/208	Remove the shutter. (Slot 1/Slot 2)
206/210	Release the tray lock after inserting the shutter. (Slot 1/Slot 2)
211	An error has occurred with the auto density measurements. Try again.
212	An error has occurred with the density measurements of the QC print. Try again.
213	The end of the operating life of the laser is close, however it can still be used. Note that the printer will need to be serviced, so contact your dealer.
214	The end of the operating life of the laser has expired. The printer will need to be serviced, so contact your dealer.
215	The film size is different. Set the tray with 14”x17” film loaded.
216	The film size is different. Set the tray with 14”x14” film loaded.
217	The film size is different. Set the tray with 10”x14” film loaded.
218	The film size is different. Set the tray with 10”x12” film loaded.
219	The film size is different. Set the tray with 8”x10” film loaded.
220	An unregistered tray has been set.Set a registered tray.
221/223	The film has become jammed in the sheet-feeder. Remove the film. (Slot 1/Slot 2)
226/227	The film has become jammed in the conveyance unit. Remove the film.
228	The film has become jammed in the recording unit. Remove the film.
230	The film has become jammed in the thermal development unit. Remove the film.
232	The film has become jammed in the discharge unit. Remove the film.
244	Reading of a barcode has failed. Load the tray again.
245	A different barcode has been detected. Load correct films.
246	Films with a different base color have been loaded. Load correct films.
249	A tray for a film size different from the setting has been loaded. Load the tray for the correct film size.
660	Contact your dealer for servicing.

If an error code other than above appears, take a note of it and contact Fujifilm service personnel.

⚠ CAUTIONS

This equipment is designed to output image data on the film using lasers. Life of the lasers is limited and managed by the equipment. When the end of their life approaches, the error code “213” appears, and about a month later, the “214” appears indicating that lasers will soon be inoperative. When the error code “214” appears, contact Fujifilm service personnel immediately.

5.5 When a Film Jam Occurred

When a film jams inside the device, an alarm sounds, an error code appears and then the processing is aborted.
Check the error code and remove the jammed film.

- **Error codes : 221, 223**..... Page **5-6**
- **Error codes : 226, 227**..... Page **5-10**
- **Error codes : 228**..... Page **5-11**
- **Error codes : 230**..... Page **5-12**
- **Error codes : 232**..... Page **5-13**

⚠CAUTIONS

Do not reload in a film tray a film that has once jammed in the equipment.
Do not take unused films out of a film tray or add new films in an already-loaded film pack.
This will result in a misoperation or failure of the equipment. Correct output result will not be obtained either.

⚠CAUTIONS

When locking a film tray or cover, be careful not to pinch your fingers or hands.

⚠CAUTIONS

You may burn yourself if your bracelet or necklace touches high-temperature parts or units, such as inlet of the Thermal Development Unit.

5.5.1 Film jamming when the error code “221” or “223” is displayed

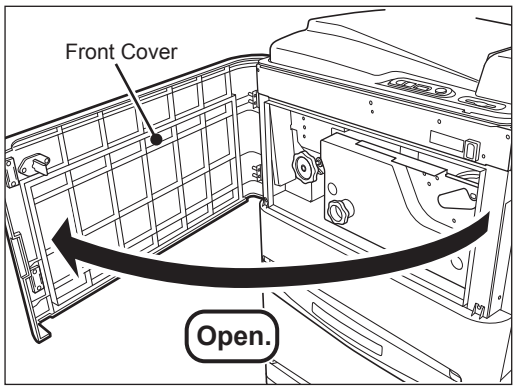
Use the procedure described below to remove a film jamming at the back of a film tray.

⚠ CAUTIONS

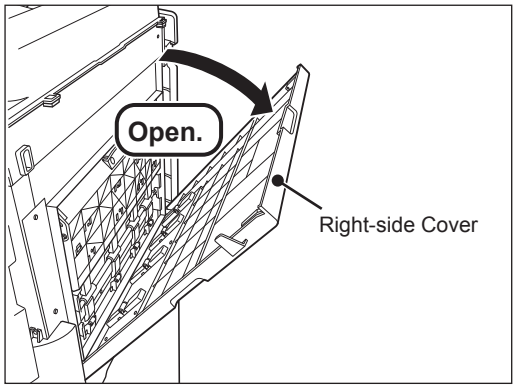
When handling any cover in the equipment, hold it by the green part. Handling a cover by holding another part may cause a malfunction or injury.

1 **Confirm that the error code “221” or “223” appears on the Display panel.**

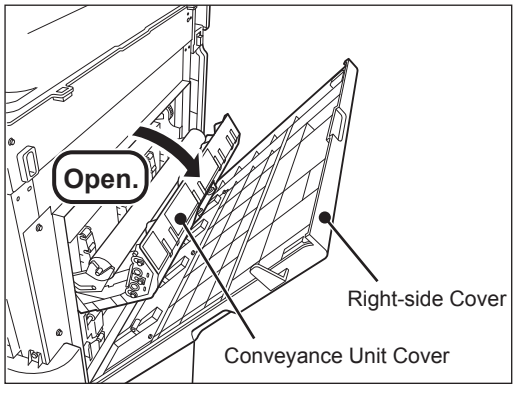
2 **Open the front cover.**



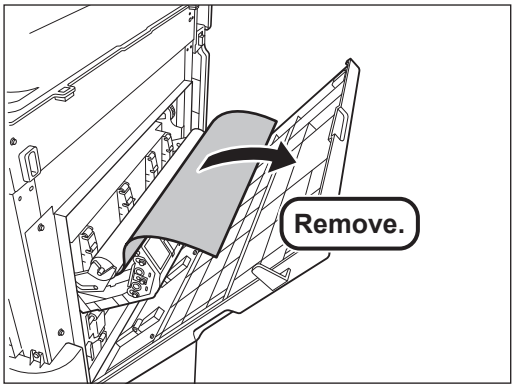
3 **Open the right-side cover.**



4 **Open the conveyance unit cover inside of the right-side cover.**



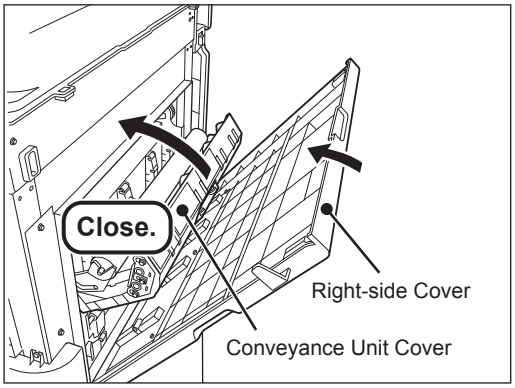
5 **Remove a jam film if there is any.**



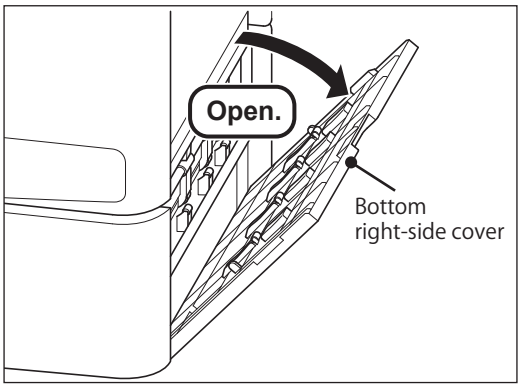
⚠ CAUTIONS

Removing a jam film forcibly may result in damaging a roller or other units.

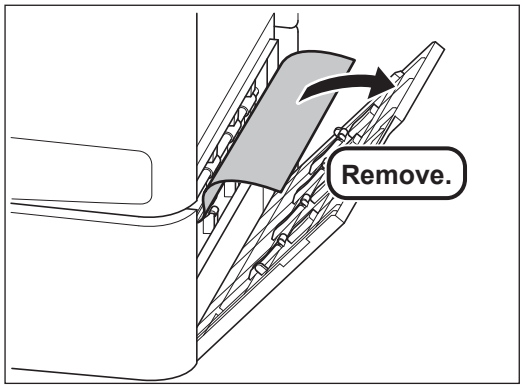
6 **Close the conveyance unit cover → right-side cover in this order.**



7 **Open the bottom right-side cover.**



8 **Remove a jam film if there is any.**

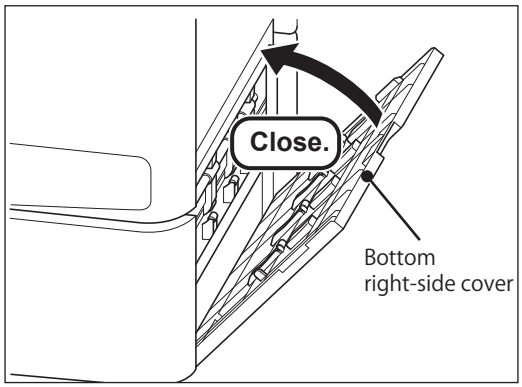


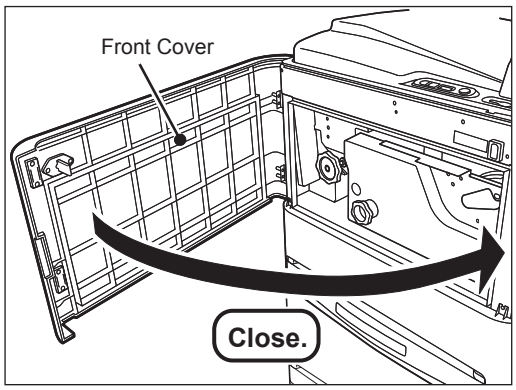
⚠ CAUTIONS

Removing a jam film forcibly may result in damaging a roller or other units.

Go to page 5-8 when there is no jam film inside the right-side cover and the bottom right-side cover

9 **Close the bottom right-side cover → front cover in this order.**





⚠ CAUTIONS

When closing covers, be careful not to pinch your fingers.

HINT

A defectively output image will be printed again.

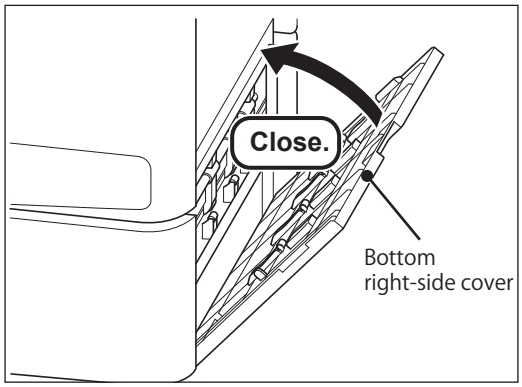
■ When there is no film jamming inside the right-side cover and the bottom right-side cover

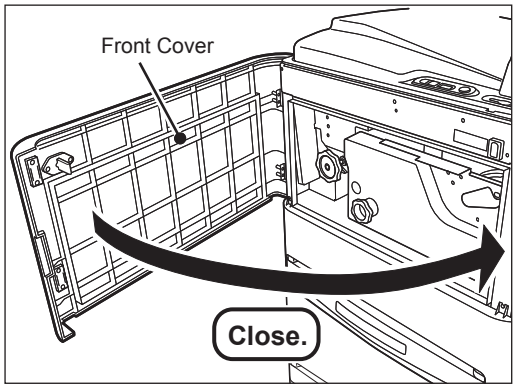
When a film is jammed at the back of the slot 1 or the slot 2, remove the jammed film in following procedures.

⚠ CAUTIONS

When handling any cover in the equipment, hold it by the green part. Handling a cover by holding another part may cause a malfunction or injury.

1 Close the bottom right-side cover → front cover in this order.

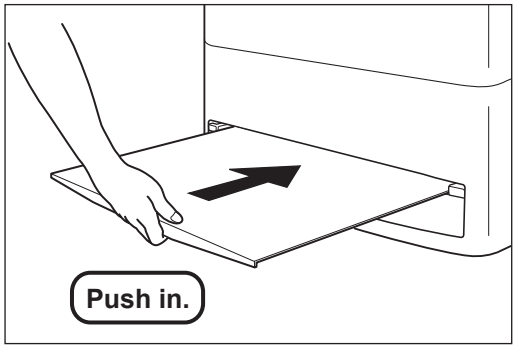




⚠ CAUTIONS

When closing covers, be careful not to pinch your fingers.

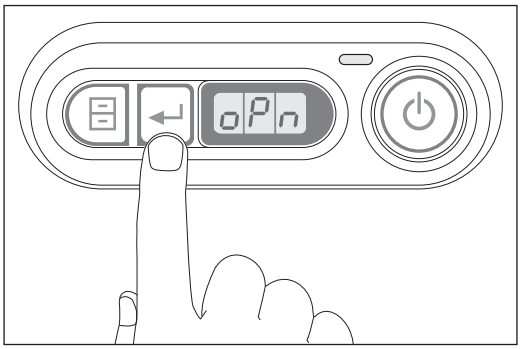
2 Carefully place the shutter in the groove on top of Film Tray 1 and push it in straight as far as it will go.



⚠ CAUTIONS

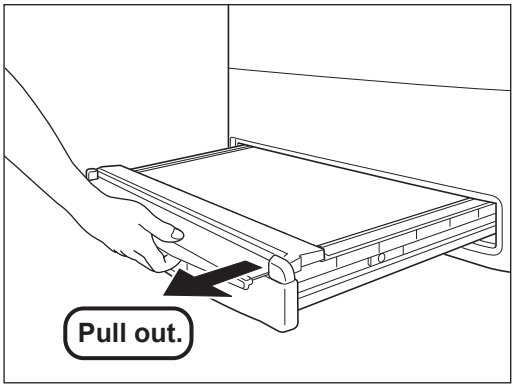
- **When inserting the shutter, be careful and hold it firmly with both hands to prevent it from dropping.**
- **When inserting or removing the shutter, take care that your hands are not caught between the parts.**
- **If the shutter cannot be inserted in the film tray, do not forcible press it but contact Fujifilm service personnel.**

3 Press the Enter Button to unlock the film tray.

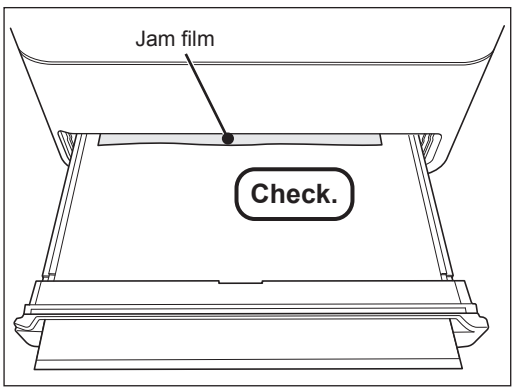


"oPn" appears on the Display panel.

4 Pull the film tray 1 out toward you.



5 Check to see if a film is jamming at the back of the film tray 1.



Go to page 5-9 if there is no film jamming at the back of the film tray.

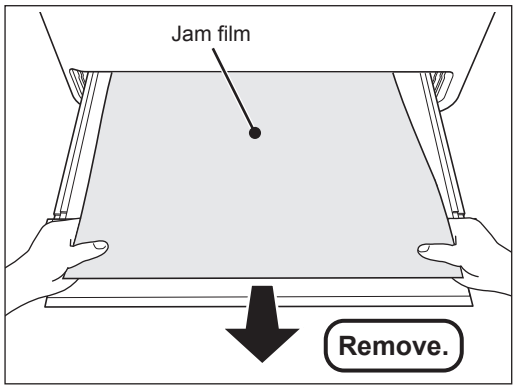
6 **If there is no film jammed at the back of film tray 1, check if there is a film jammed at the back of film tray 2.**

CAUTIONS

To unlock the film tray 2, insert the shutter only in the film tray 2.
The film tray 1 will be unlocked when the shutters are inserted into both the film tray 1 and the film tray 2.

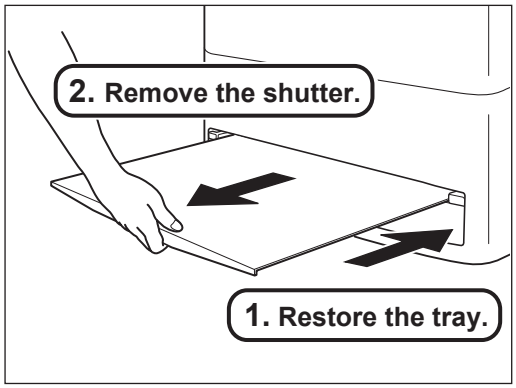
Go to page 5-9 when there is no jam film at the back of film tray 1 and film tray 2.

7 **Carefully pull the jam film out toward you to remove it.**



8 **Push the film tray in firmly as far as it will go (1.), and remove then the shutter (2.).**

After 3 to 5 minutes approximately, the equipment will be ready for printing.



CAUTIONS

- **When closing covers, be careful not to pinch your fingers.**
- **When inserting or removing the shutter, take care that your hands are not caught between the parts.**
- **When removing the shutter, be careful and hold it firmly with both hands to prevent it from dropping.**

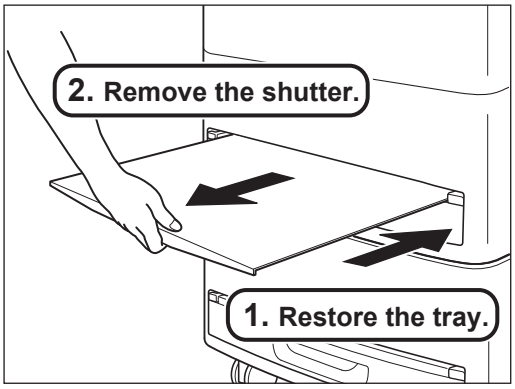
HINT

A defectively output image will be printed again.

■ When there is no film jamming at the back of a film tray

1 **Push the film tray in firmly as far as it will go (1.), and remove then the shutter (2.).**

After 3 to 5 minutes approximately, the equipment will be ready for printing.



If the error code reappears, hold down the Stand-by switch for a while to shut down the equipment and restart it. If the error code still reappears, contact Fujifilm service personnel.

HINT

A defectively output image will be printed again.

CAUTIONS

- **When inserting or removing the shutter, take care that your hands are not caught between the parts.**
- **When removing the shutter, be careful and hold it firmly with both hands to prevent it from dropping.**

5.5.2 Film jamming when the error code “226” or “227” is displayed

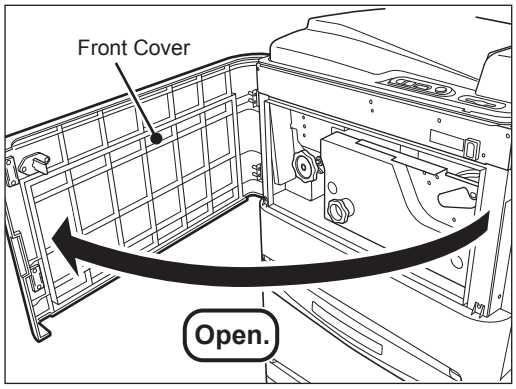
Use the procedure described below to remove a film jamming in the Conveyance Unit.

⚠ CAUTIONS

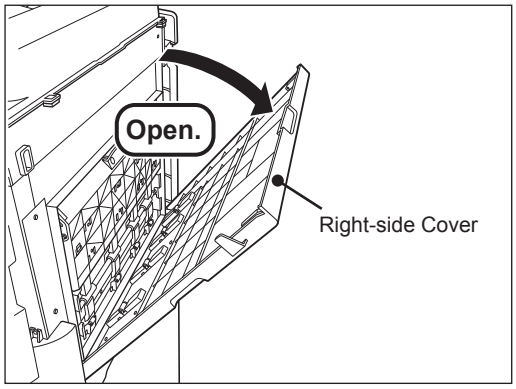
When handling any cover in the equipment, hold it by the green part. Handling a cover by holding another part may cause a malfunction or injury.

1 Confirm that the error code “226” or “227” appears on the Display panel.

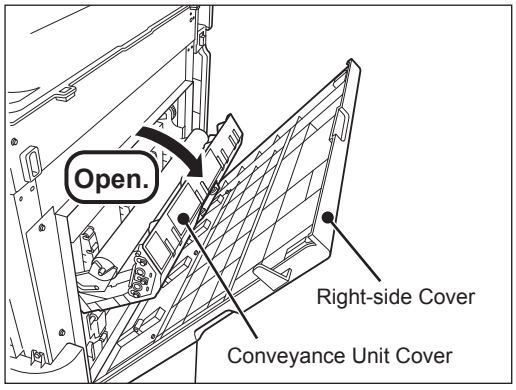
2 Open the front cover.



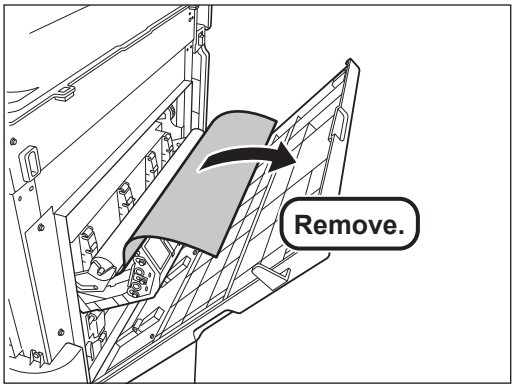
3 Open the right-side cover.



4 Open the conveyance unit cover inside of the right-side cover.

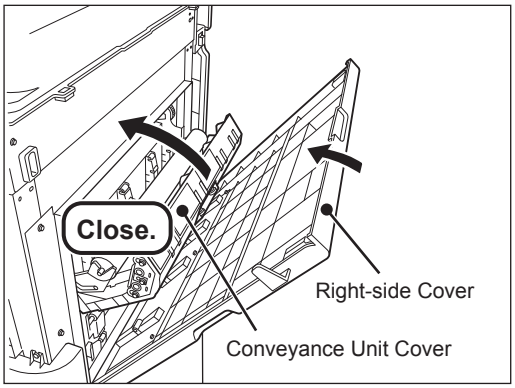


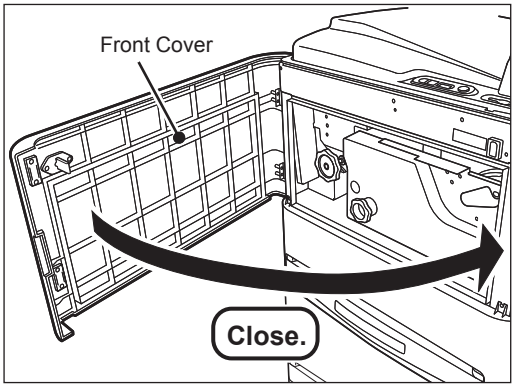
5 Remove a jam film, if there is any.



6 Close the conveyance unit cover → right-side cover → front cover in this order.

After 3 to 5 minutes approximately, the equipment will be ready for printing.





⚠ CAUTIONS

When closing covers, be careful not to pinch your fingers.

HINT

A defectively output image will be printed again.

5.5.3 Film jamming when the error code “228” is displayed

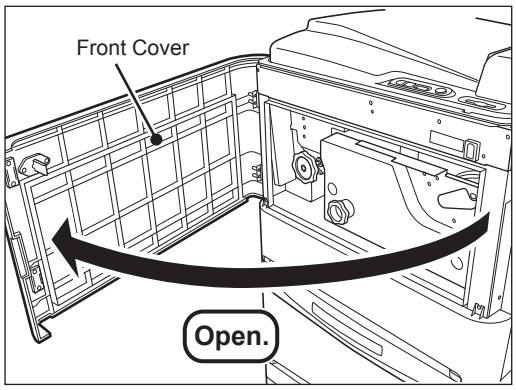
Use the procedure described below to remove a film jamming in the Recording Unit.

CAUTIONS

When handling any cover in the equipment, hold it by the green part. Handling a cover by holding another part may cause a malfunction or injury.

1 Confirm that the error code “228” appears on the Display panel.

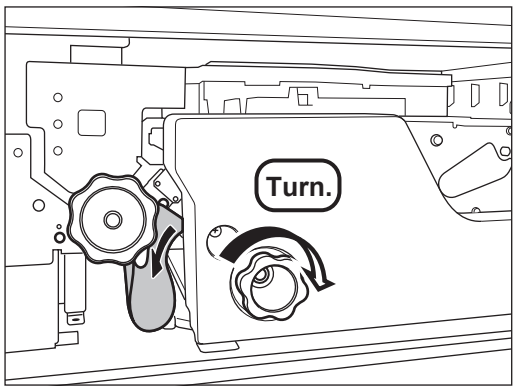
2 Open the front cover.



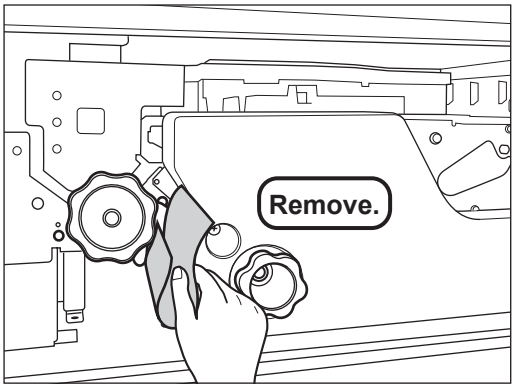
3 Slowly turn the right-side handle clockwise to feed the jammed film.

HINT

If the handle is turned when power is not supplied to the equipment, the pilot lamp on the Operation Panel may turn on. This does not indicate a functional or performance problem.

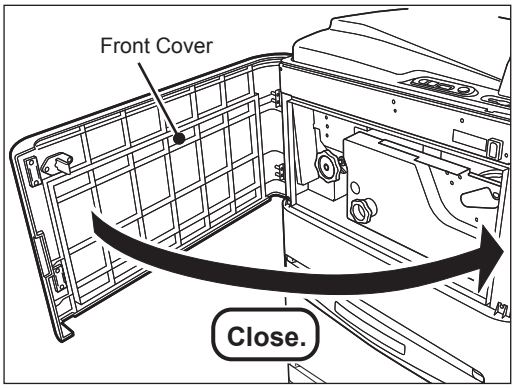


4 Remove then the jam film.



5 Close the front cover firmly.

After 3 to 5 minutes approximately, the equipment will be ready for printing.



CAUTIONS

When closing covers, be careful not to pinch your fingers.

HINT

A defectively output image will be printed again.

5.5.4 Film jamming when the error code “230” is displayed

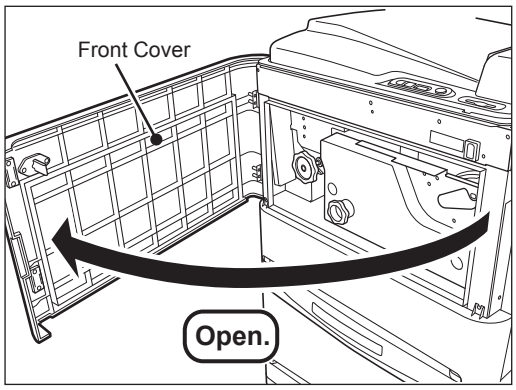
Use the procedure described below to remove a film jamming in the Thermal Development Unit.

CAUTIONS

When handling any cover in the equipment, hold it by the green part. Handling a cover by holding another part may cause a malfunction or injury.

1 Confirm that the error code “230” appears on the Display panel.

2 Open the front cover.

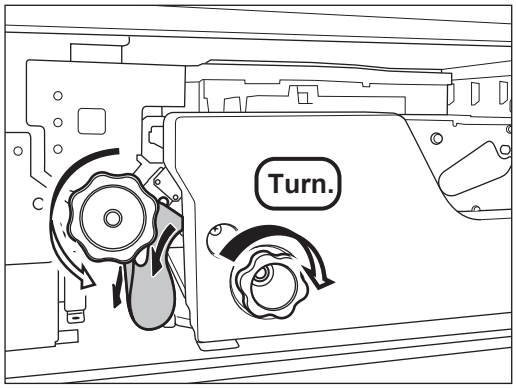


3 Slowly turn the right-side and left-side handles to feed the jammed film.

Turn the two handles one by one until a film end comes out completely.

HINT

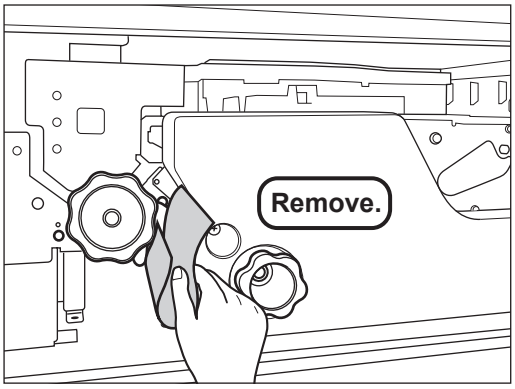
If the handle is turned when power is not supplied to the equipment, the pilot lamp on the Operation Panel may turn on. This does not indicate a functional or performance problem.



HINT

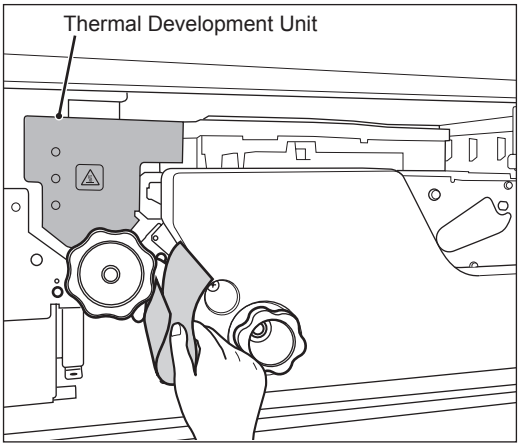
- Turn the left-side handle counterclockwise and the right-side handle, clockwise.
- If the film is hidden, turn the left-side handle clockwise. Remove the film ejected from the ejection unit.

4 Remove the jam film.



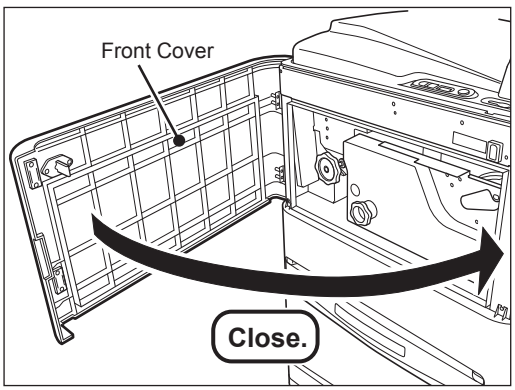
CAUTIONS

Be careful not to touch the Thermal Development Unit.



5 Close the front cover firmly.

After 3 to 5 minutes approximately, the equipment will be ready for printing.



CAUTIONS

When closing the cover, be careful not to pinch your fingers.

HINT

A defectively output image will be printed again.

5.5.5 Film jamming when the error code “232” is displayed

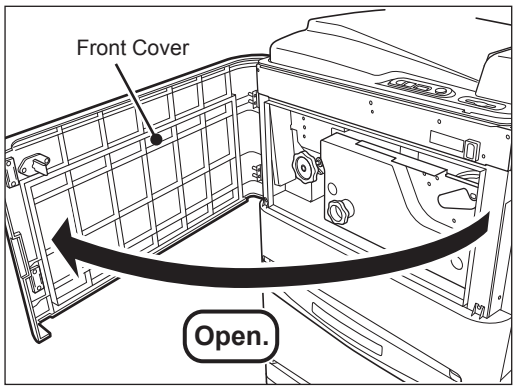
Use the procedure described below to remove a film jamming in the Ejection Unit.

CAUTIONS

When handling any cover in the equipment, hold it by the green part. Handling a cover by holding another part may cause a malfunction or injury.

1 Confirm that the error codes “232” appear on the display panel.

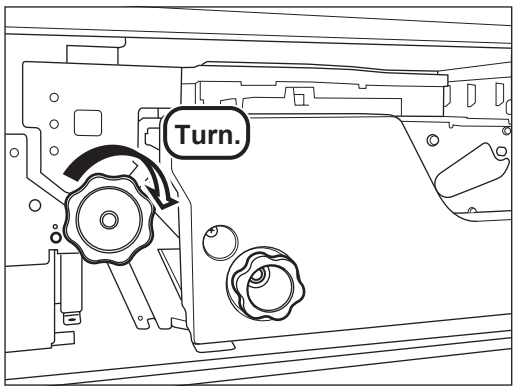
2 Open the front cover.



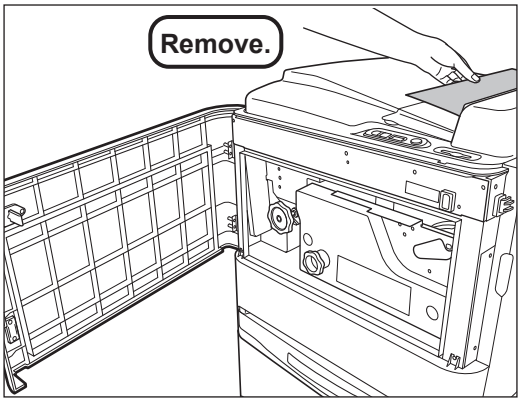
3 Slowly turn the left-side handle clockwise to feed the jammed film.

HINT

If the handle is turned when power is not supplied to the equipment, the pilot lamp on the Operation Panel may turn on. This does not indicate a functional or performance problem.

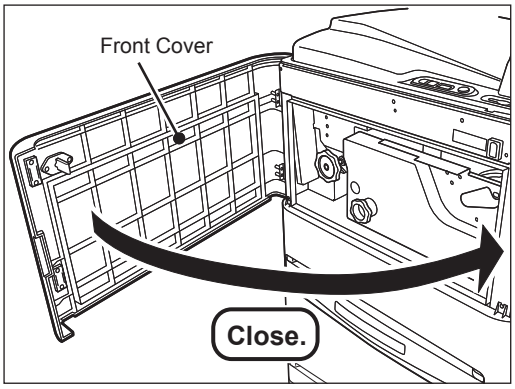


4 Remove then the jam film.



5 Close the front cover firmly.

After 3 to 5 minutes approximately, the equipment will be ready for printing.



CAUTIONS

When closing covers, be careful not to pinch your fingers.

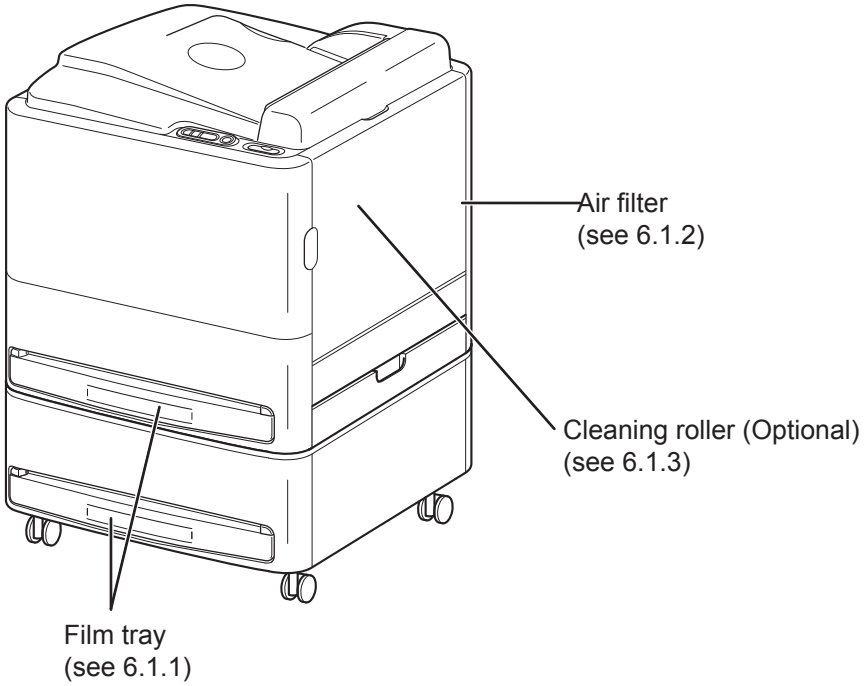
HINT

A defectively output image will be printed again.

Chapter 6 Care and Maintenance

6.1 Cleaning the Inside and Surroundings of the Equipment

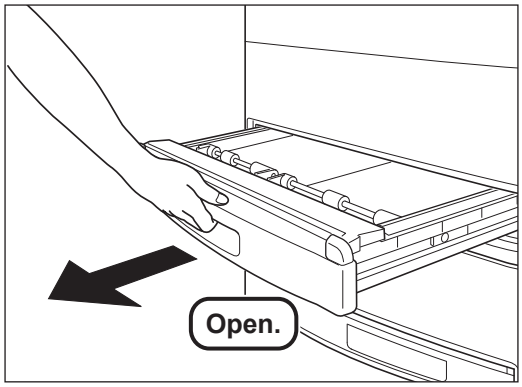
Once every 3 months, clean the equipment by removing dirt or dust on it.



6.1.1 Cleaning a film tray

Use of dirty film may lead to the degradation of images or equipment malfunction. To prevent this, clean the film tray when replacing a film pack or on other occasions.

1 **Pull the film tray out toward you.**



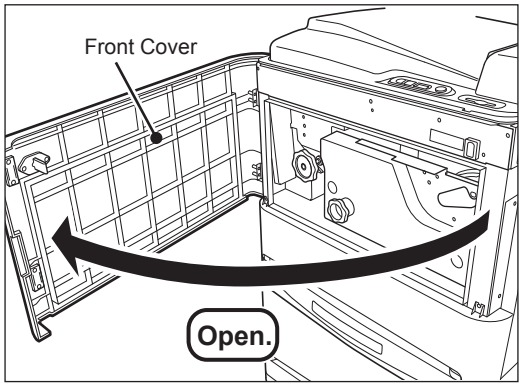
2 **Vacuum dirt or dust in the film tray.**

⚠ CAUTIONS

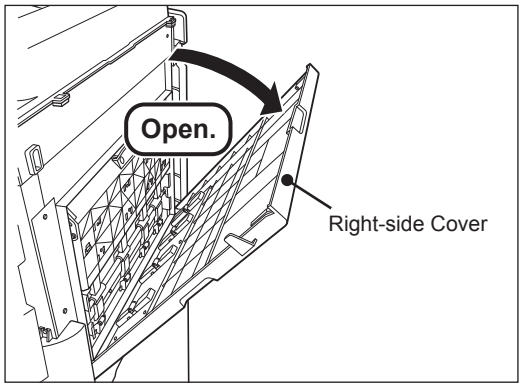
Do not use ethanol.
If ethanol is impregnated into the barcode reader filter, it can be damaged.

6.1.2 Cleaning the Air Filter

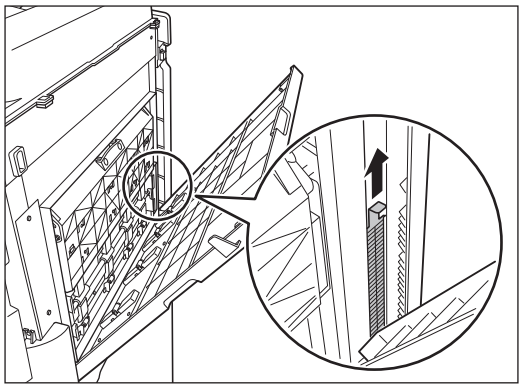
1 **Open the front cover.**



2 **Open the right-side cover.**

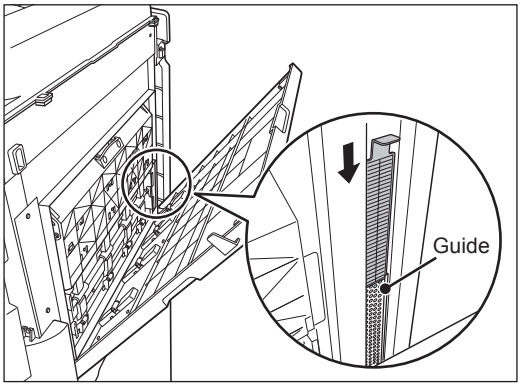


3 **Lift and remove the air filter at the rear of the right-side cover by holding the locking tab.**

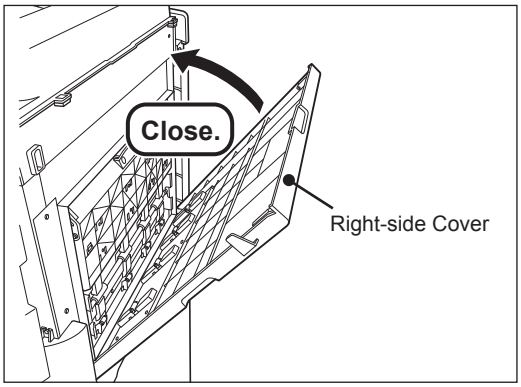


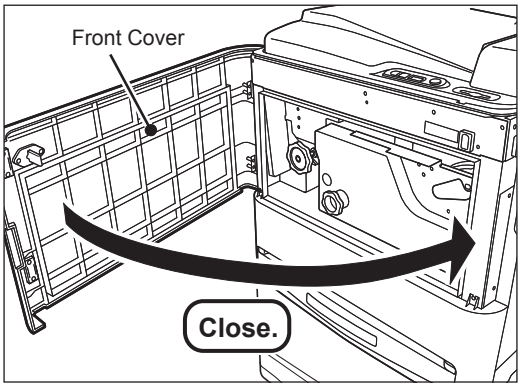
4 **Vacuum dust adhered to the filter.**

5 **Install the air filter by holding the locking tab.**



6 **Close the right-side cover → front cover in this order.**





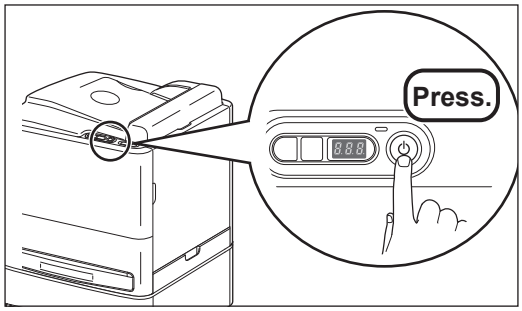
6.1.3 Washing the Cleaning Roller (Optional) with Water

We recommend that you wash the cleaning roller with water once every week to secure print image quality.
When white spots or traces of lint appear on output films, also wash the cleaning roller with water following the procedure described herein.

⚠ CAUTIONS

Do not use the following procedure to clean parts other than the cleaning rollers.

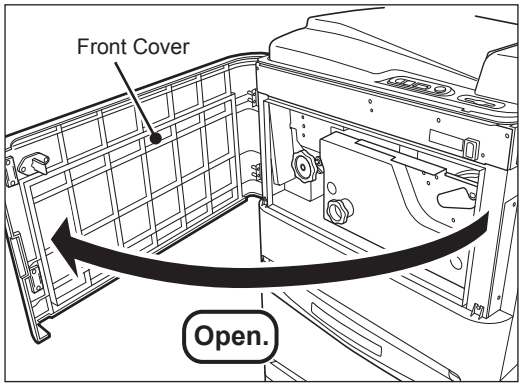
1 Make sure that no film recording or output processing is being performed, and then shut the system down.



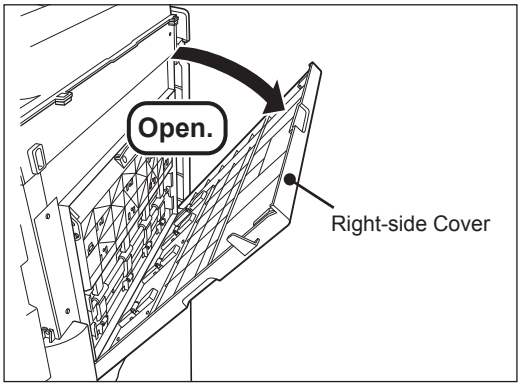
⚠ CAUTIONS

An error may occur if procedural steps that follow are performed without shutting down the system.

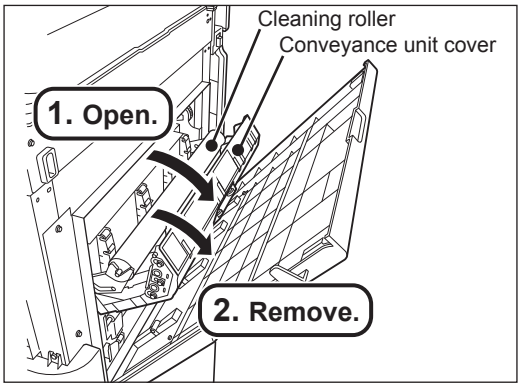
2 Open the front cover.



3 Open the right-side cover.



4 Open the conveyance unit cover (1.) to remove the cleaning roller (2.).

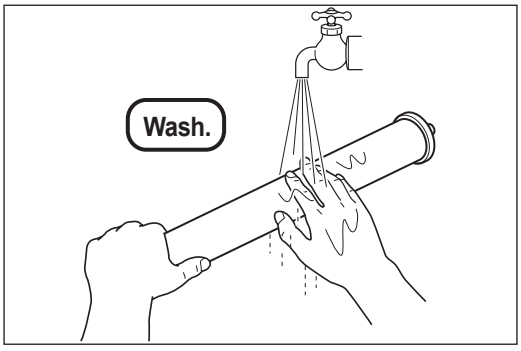


⚠ CAUTIONS

When removing the cleaning roller, hold it with both hands being careful not to drop it on the floor.

5 Wash the cleaning roller with water.

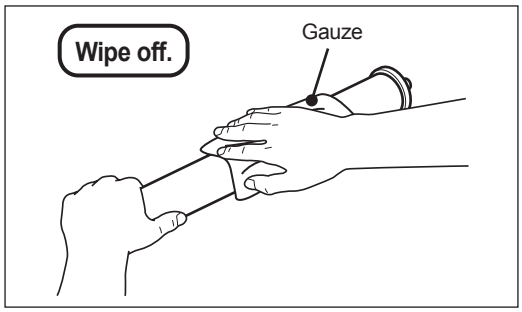
While splashing water onto the cleaning roller, rub the surface gently to clean.



⚠ CAUTIONS

If the cleaning roller is rubbed roughly to removing stubborn dirt or dust adhered to the surface and that does not come off only with water, the roller surface may be damaged.
If white spots still appear on output films even after it has been water-washed, it is time to replace the cleaning roller for a new one.
Replace the cleaning roller every three years.

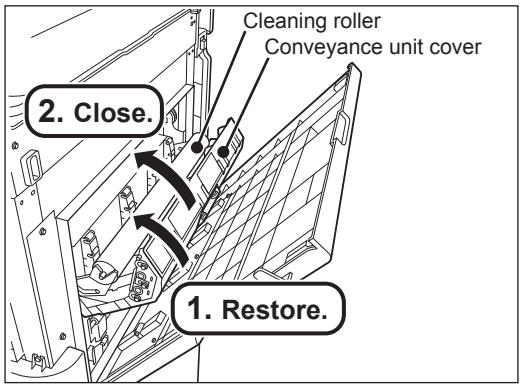
6 **Use a gauze moistened with alcohol (ethanol anhydride) to wipe off water remaining on the cleaning roller. (The cleaning roller will be dried in a few minutes.)**



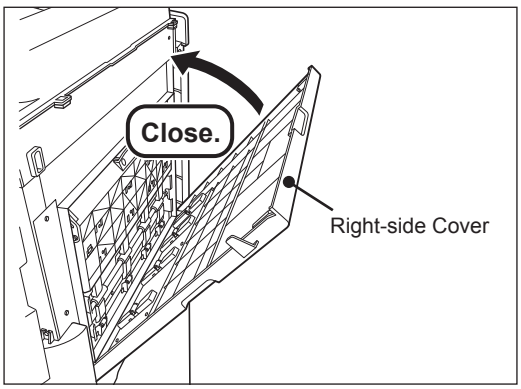
⚠ CAUTIONS

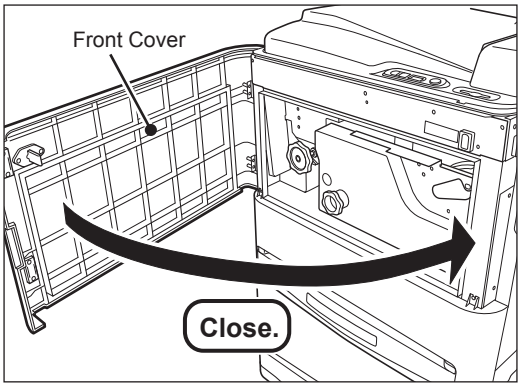
Dry the cleaning roller sufficiently, otherwise correct film density may not be obtained or unevenness may appear on an output film. Water does not enter the device.

7 **Restore the cleaning roller (1.), and close the conveyance unit cover (2.).**



8 **Close the right-side cover → front cover in this order.**





6.1.4 Image Verification

A white blurred streak appears vertically on the printed image, as shown below.

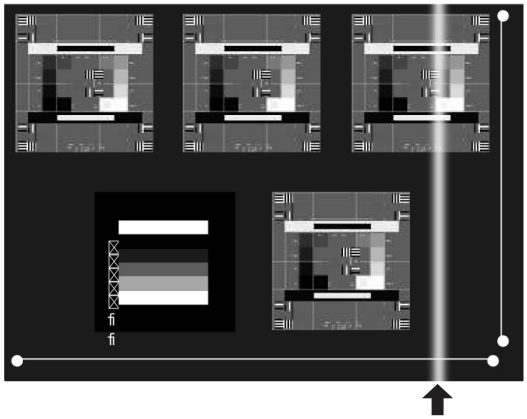
1 **Print a test pattern (QC test pattern).**

For details of printing a QC test pattern, see “4.5 Outputting a QC Test Pattern” under Chapter 4.

2 **Verify the test pattern.**

If a similar streak appears on the printed test pattern film, it is possible that the DRYPIX Smart optical unit is soiled. In such cases, contact Fujifilm service personnel.

<26 × 36cm>



(On this photo, the white streak is shown emphasized so that it is visible clearly.)

6.2 Users Checksheet (Care and Maintenance)

We recommend that you periodically check the following so that you can optimally operate the equipment constantly. Make a copy when you are using this checksheet.

● **Verification**

Details	Frequency	Results of users verification
Are the power cable and/or grounding cable connected/fastened properly?	Daily	
Is the cable connected/ fastened properly between this equipment and other equipment?	Daily	

● **Cleaning**

Unit for cleaning	Frequency	Reference	Date of cleaning
Film Tray	Every three months	6-2	
Air Filter	Every three months	6-3	
Cleaning Roller (Optional)	Weekly	6-4	
Image verification	Every six months	6-6	

6.3 About Preventive Maintenance

To maintain safety for the customer, the user, and other people, perform the preventive maintenance mainly involved cleaning of the parts and replacing of consumables. For the preventive maintenance, consult our official dealer or local representative.
Necessary cycle of periodical maintenance or part replacement depend on the usage conditions or operation time per day.
Details of maintenance or inspections performed by the specified dealers differ depending on the maintenance contract.

⚠ CAUTIONS

The life of this equipment is six years.
To use the equipment after this period, contact a designated service supplier to perform an overhaul.
For maintenance and inspection after that, consult with a designated service supplier.

■ Preventive maintenance to be performed when the machine has been used for one year.

Perform the preventive maintenance shown below.

Cleaning of the parts
- Cleaning roller (optional)
- Removal unit
- Covers
- Inside the equipment
- Rubber belt
- Sub-scanning conveyance roller
- Density measurement section

Checking the Machine
- Image/conveyance
- Interlock function(Front cover/Right cover/Bottom right-side cover/Film tray)
- Tray periphery
- Error log
- Setting date and time
- Rubber belt
- Improper protective grounding
- Final operation

Replacing of consumables
- Film cutter

■ Preventive maintenance to be performed when the machine has been used for two years

Perform the preventive maintenance shown below.

Replacing of consumables
- Air filter

■ Preventive maintenance to be performed when the machine has been used for three years

Perform the preventive maintenance shown below.

Replacing of consumables
- Cleaning roller (optional)

Cleaning of the parts
- Heat development unit

Appendix A Specifications

A.1 Specifications

The specifications of the DRYPIX Smart are shown below.

Power Supply Conditions

Input voltage	: North America AC100/110/120V(Rated current:10.6/11.2/12A) Other countries AC200/220/230/240V(Rated current:5.3/5.6/5.9/6A)
Frequency	: 50Hz-60Hz
Allowable variations in voltage	: ± 10%
Phase	: Single

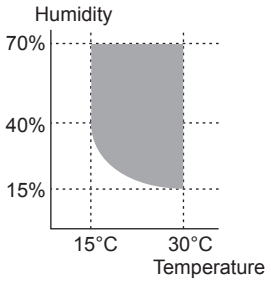
Environmental Conditions

(1) Operating conditions

Temperature/humidity	: 15ºC (40 to 70%RH) to 30ºC (15 to 70%RH) (no dew condensation)
Atmospheric pressure	: 750hPa to 1060hPa
Maximum heating value	: 1300kJ

(2) Non-operating conditions

Temperature	: 0ºC to 45ºC
Humidity	: 10%RH to 90%RH (no dew condensation)
Atmospheric pressure	: 750hPa to 1060hPa



Supplies

Fuji Medical Dry Imaging Film : DI-HL (Blue base), DI-ML (Blue base)
Film cutter

Options

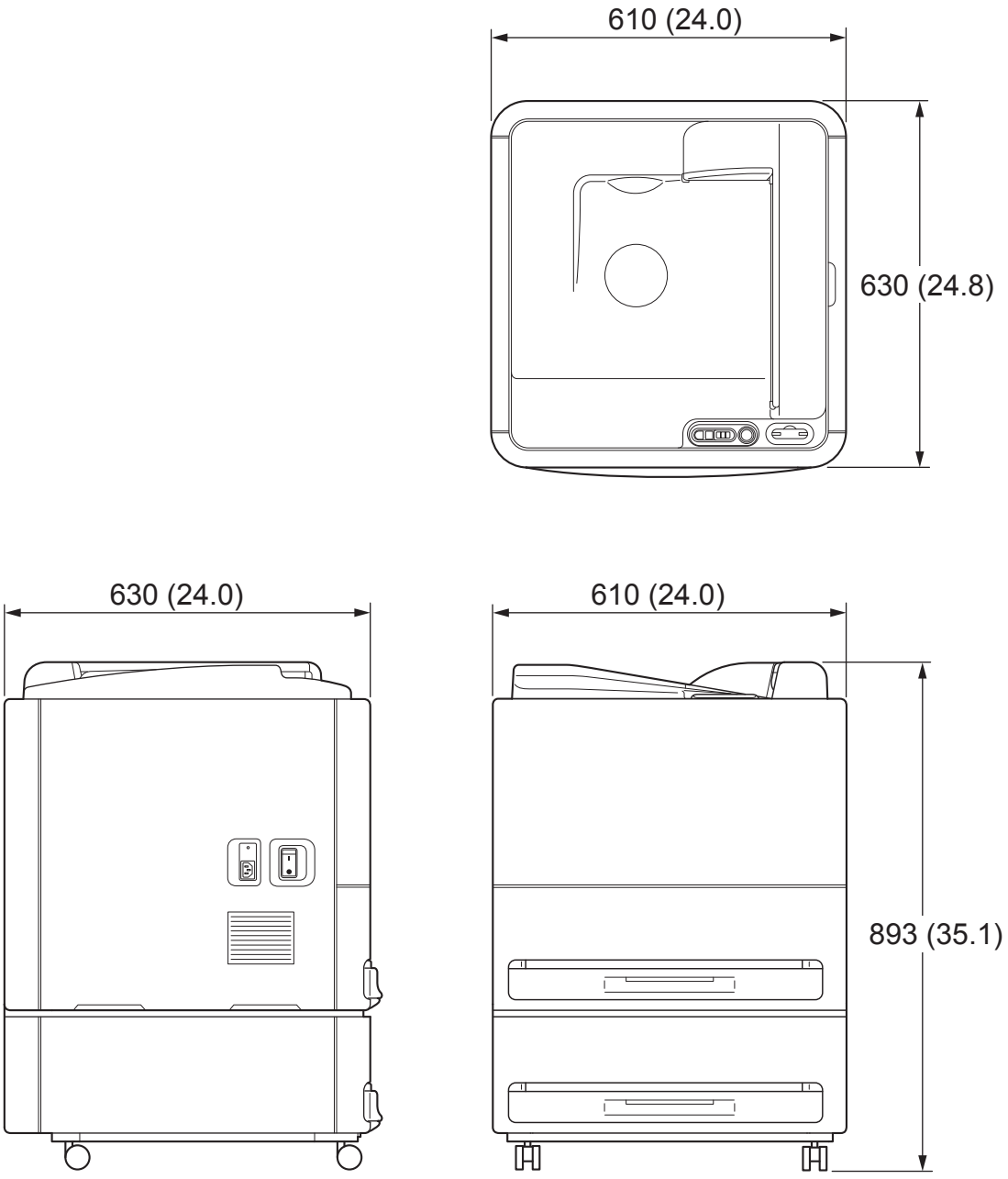
Film tray, Power cable, Cleaning roller, Fxture and Additional suction cup kit

A.2 External View and Weight

The dimensions and weight of the DRYPIX Smart are shown below.

Model Name	Width (mm (in.))	Depth (mm (in.))	Height (mm (in.))	Weight (kg (lb))
DRYPIX Smart	Approx. 610 (24.0)	Approx. 630 (24.8)	Approx. 893 (35.1)	Approx. 104 (229.3)

(Unit : mm (in.))



Maintenance and Inspection

1 Maintenance and Inspection Items Assigned to Specified Dealer

For periodical inspection of the equipment and necessary arrangements, consult our official dealer or local representative.

Manufacturer :

FUJIFILM Corporation

26-30, NISHIAZABU 2-CHOME, MINATO-KU, TOKYO 106-8620, JAPAN

CE

EC REP European Authorized Representative:
FUJIFILM Europe GmbH
Heesenstrasse 31, 40549 Duesseldorf, Germany

EU Importer:
FUJIFILM Europe B.V.
Oudenstaart 1, 5047 TK Tilburg, The Netherlands

FUJIFILM MEDICAL SYSTEMS U.S.A., INC.

419 WEST AVENUE, STAMFORD CT 06902, U.S.A.